BanglaFactGuard

A Multimodal Bangla Fact-Checking System

Final Report
SE 801: Software Project Lab III

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Submission Date: 11 October 2026

Institute of Information Technology
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Letter of Transmittal

11 October 2026

BSSE 4th Year Exam Committee

Institute of Information Technology

University of Dhaka

Subject: Submission of the BanglaFactGuard final report

Dear Sir,

I am submitting the final report for BanglaFactGuard, my Software Project Lab III project. The report explains the requirements, system design, implementation and testing. It also includes a user manual and instructions for setting up the project.

The report describes the work completed and the limitations that remain. I have included the design diagrams, interface screenshots and recorded test results to support the discussion. I would appreciate your feedback on the project and this report.

Thank you for your consideration.

Sincerely,

Md. Abul Kalam Azad

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_____

Supervisor's Signature

Toukir Ahammed

Lecturer

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Acknowledgement

I am sincerely grateful to the Institute of Information Technology, University of Dhaka, for giving me the opportunity to carry out my Software Project Lab III on the project titled "BanglaFactGuard: A Multimodal Fact Checking System". This project has given me the chance to apply theoretical knowledge of software engineering, natural language processing, computer vision and information retrieval to a practical, socially relevant problem.

I would like to express my heartfelt gratitude to my respected supervisor, Toukir Ahammed, Lecturer, Institute of Information Technology, University of Dhaka, for his continuous guidance, constructive criticism and encouragement throughout every phase of this work. His insights into multimodal machine learning, Bangla natural language processing and system design have been instrumental in shaping the direction, design and scope of this system.

I am also thankful to my other faculties and classmates for their cooperation, helpful discussions and suggestions during different phases of the project. Finally, I acknowledge the resources and academic environment provided by the Institute of Information Technology, which made the successful progress of this project possible.

Abstract

Bangla news claims and photocards can contain altered headlines, misleading source information or incorrect dates. Checking this content manually requires users to find the original report and compare its details. BanglaFactGuard aims to support this process by bringing automated analysis, published evidence and expert review into one platform.

The platform uses Angular for the web application, FastAPI for backend services and a Chrome extension for browser access. PostgreSQL stores application data, Redis supports caching and MinIO stores uploaded images. Gemini extracts information from photocards, while BanglaBERT and EfficientNet-B4 support text-and-image analysis. Background jobs handle long-running verification tasks.

BanglaFactGuard supports text verification, photocard verification and multimodal text-and-image analysis. Selecting a news source is optional: the system searches the selected publisher when one is provided or registered active news sources otherwise. It assesses source correspondence, headline alteration and publication-date consistency separately. Photocard verification extracts the printed headline, outlet and date before checking the available evidence. Additional features include user authentication, submission history, evidence reports, notifications, expert review and administrative oversight. Automated assessments remain separate from expert-finalized verdicts and missing evidence is not treated as proof that a claim is false.

Automated tests covered the backend, web application and browser extension. The recorded test run completed with 424 tests passed and two skipped. Separately, the headline comparison model achieved 80% accuracy on its held-out test set. This score measures the model's performance rather than the accuracy of the complete verification workflow, which also depends on extraction quality and evidence availability.

Future work should focus on evaluating complete verification outcomes, improving extraction from difficult photocards, expanding evidence coverage and reducing processing time. BanglaFactGuard provides a practical foundation for investigating Bangla news content by presenting automated findings alongside supporting evidence and opportunities for human review..

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Project Title: BanglaFactGuard: A Multimodal Bangla Fact-Checking System

GitHub Repository: https://github.com/abulkalamazad1425/BanglaFactGuard

1. Introduction

Social media has made it easy to share news and information quickly. However, it has also made it easier for fake news and rumours to spread. In Bangladesh, misleading information is often shared through Bangla posts, screenshots and news photocards. Some fake content uses the logos and designs of well-known news organisations, making it difficult for people to identify fake news. Technology and AI can help people check news by finding related articles and collecting useful information. However, AI predictions are not always correct, so human review is also important. BanglaFactGuard is a news verification system that helps users check suspicious Bangla news. Users can submit news text or images, search previously checked claims and view supporting evidence. The system provides an initial assessment and experts review the evidence before giving the final verdict. This report describes the project's requirements, design, development, testing and results.

1.1 Problem Statement

People often share news on social media without checking whether it is true. Verifying a news claim requires finding the original article, checking its publication date, comparing the headline and looking for reliable evidence. These steps take time and many users do not know how to verify news properly.

Fake news may also use the logos of trusted news organisations or old images to make misleading claims look real. Although fact-checking services help investigate such claims, finding the right report can be difficult and checking new claims takes time. AI-based systems can also make mistakes when they cannot find the correct article or enough evidence.

As a result, misleading news may spread before it is properly checked. BanglaFactGuard aims to reduce this problem by allowing users to submit suspicious news, find previous verification results and view related evidence. It uses AI for an initial assessment and expert review for the final verdict, helping users check Bangla news more easily and reliably.

1.2 Objectives

The objectives of BanglaFactGuard are:

- To let users check Bangla news claims in three ways: Text & source, Photo card and Text & image.
- To compare claims with relevant news reports and present separate findings for the source, headline and publication date.
- To provide a preliminary prediction for news submitted as text and an image.
- To support expert voting and administrator decisions for claims that cannot be resolved within the review limits.
- To reuse suitable results from earlier checks and identify when a result has been reused.
- To provide a public Fact Explorer, personal submission history and result notifications.

- To provide a browser extension so users can check a claim from the page they are reading.
- To provide tools for managing experts, registered news sources and voting rules.

1.3 Scope

1.3.1 In Scope

- A website that supports guests, registered users, experts and administrators.
- Three verification methods: Text & source, Photo card and Text & image.
- Separate source, headline and date findings, with four similarity measures when article text is supplied.
- Background processing, progress updates and reuse of compatible saved results.
- Expert review, credibility-weighted voting, escalation and administrator decisions.
- Public Fact Explorer with search, filters and statistics.
- In-app notifications for signed-in submitters and queued final-result emails.
- A Chrome extension with text import, image upload, selected-area capture, cropping, activity history and alerts.
- Administration of experts, verified news sources, credibility tiers and voting configuration.

1.3.2 Out of Scope

- Checking claims in languages other than Bangla.
- Native mobile applications; the system is a website and a desktop Chrome extension.
- Video verification.
- Fully automated final verdicts; every final verdict comes from experts or an administrator.
- Automatic crawling of social media; only submitted claims are checked.
- Removing content or taking legal action against publishers.

1.4 Deliverables

- The BanglaFactGuard website, backend service and Chrome extension.
- The training notebook and the trained model files required for text and image prediction.
- The database design, database update scripts and source registration setup.
- Automated test suites for the backend, the website and the extension.
- This final report, with design, implementation, testing, a user manual and installation steps.
- A public Git repository: https://github.com/abulkalamazad1425/BanglaFactGuard.
- The extension is installed from its build folder as an unpacked Chrome extension. Section 11.6 gives the installation steps.

1.5 Functional and Non-Functional Requirements

1.5.1 Functional Requirements

Table 1.1 summarises the main functions supported by the system.

Table 1.1: Functional Requirements

ID	Requirement	Users
FR-1	Create an account, sign in and out and reset a forgotten password using an emailed verification code.	Registered user, Expert & Admin
FR-2	Submit a Bangla headline, with an optional news outlet, article body and claimed date. Use registered sources when no active outlet is identified.	All users
FR-3	Read a headline and any visible outlet or date from a photo card. Explain when the image cannot be processed.	All users
FR-4	Submit a headline (oprtional), article text and image to receive a preliminary Likely Real or Likely Fake prediction.	All users
FR-5	Present source, headline and date findings separately. Explain when a comparison is unavailable or incomplete.	All users
FR-6	Show body-similarity when article text is supplied.	All users
FR-7	Reuse a compatible completed result and tell the user when this occurs.	All users
FR-8	Process accepted claims in the background and display their progress.	All users
FR-9	Allow experts to inspect claims and evidence, then submit a verdict with a justification of at least 50 characters.	Expert
FR-10	Apply the configured voting rules and send unresolved claims to an administrator when a review limit is reached.	Expert, Administrator
FR-11	Allow administrators to decide escalated claims and inspect other claims without changing their verdicts.	Administrator
FR-12	Provide a public Fact Explorer with search, filters and statistics.	All users
FR-13	Provide personal submission history and notifications, with final-result email delivery when configured.	Registered User, Expert, Administrator
FR-14	Manage expert accounts, verified sources, credibility tiers and voting configuration.	Administrator
FR-15	Offer all three methods in the browser extension, with selected-text import, screenshot capture, cropping, activity and alerts.	All users

1.5.2 Non-Functional Requirements

Table 1.2 describes the main quality requirements and the mechanisms used to address them. Performance and accuracy limits are discussed in Chapter 9.

Table 1.2: Non-Functional Requirements

ID	Category	Requirement and supporting mechanism
NFR-1	Performance	Long checks run in the background; the website and extension stay usable and show the current status.
NFR-2	Efficiency	Reuse compatible results and cache search data and text representations to reduce repeated processing.
NFR-3	Security	Store hashed passwords and use expiring access tokens. Password reset codes expire and can be used only once.
NFR-4	Security	Restrict expert and administrator actions by role, protect personal records and prevent experts from reviewing their own claims.
NFR-5	Usability	Accept Bangla content and explain findings through readable labels, result details and a public FAQ.
NFR-6	Reliability	Keep accepted jobs in the database, recover abandoned work and show a reason when processing fails.
NFR-7	Data integrity	Save a submission and its job together, avoid repeated result records and retain the weight applied to each vote.
NFR-8	Correctness and transparency	Keep automated findings separate from final verdicts. Do not label a claim Fake solely because no source report was found.
NFR-9	Maintainability	Separate features into modules and allow administrators to update sources and voting rules through the interface.
NFR-10	Compatibility	Support the Chrome extension and website with the same accounts while keeping their login sessions separate.

2. Activity Diagrams

The activity diagrams describe the main user workflows, from submitting a claim to receiving a reviewed result. Each explanation identifies the input, the main decision points and the outcome. Account settings, notifications and source management are covered in Chapters 7 and 10.

2.1 Authentication and Registration

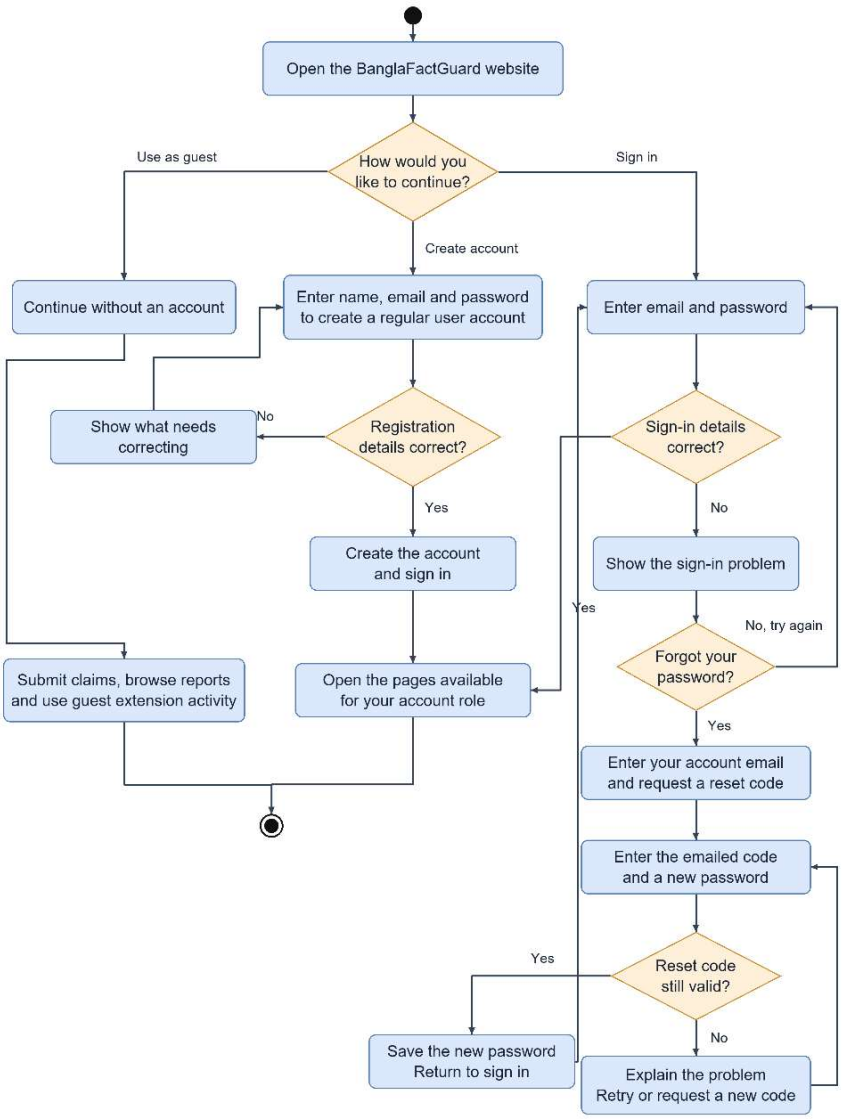


Figure 2.1: Activity Diagram — Authentication and Registration

Visitors can continue as guests, register or sign in. Registration checks the submitted details before creating an account. If sign-in fails, the form displays an error. Password recovery uses an emailed code and a new password. After sign-in, the available pages depend on the account role.

2.2 Text & Source Verification

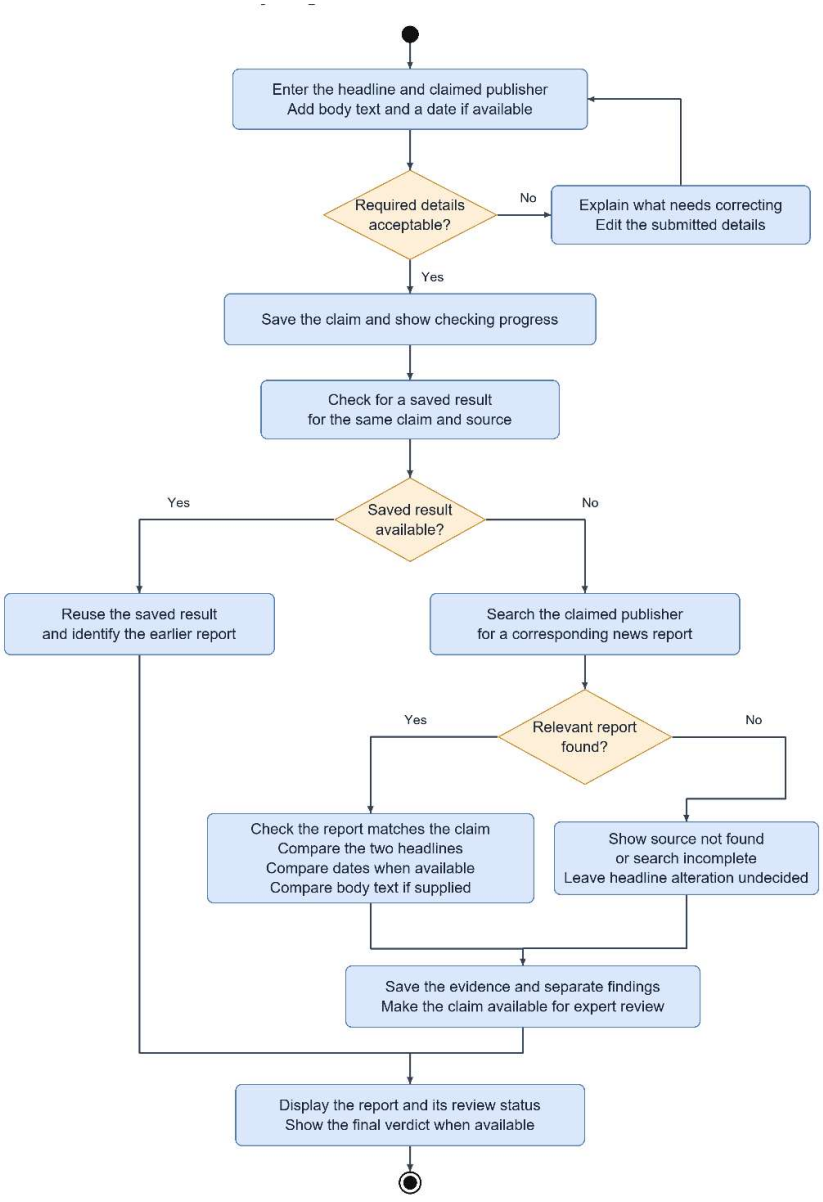


Figure 2.2: Activity Diagram — Text & Source Verification

The user submits a headline and may add the outlet, article text and date. The system reuses a compatible result or searches for relevant reports. A recognised outlet limits the search to that publisher; otherwise, the current system can search active registered sources. It then reports the available comparisons and sends the claim for expert review. An unsuccessful or incomplete search does not produce a Fake verdict.

2.3 Photo Card Verification

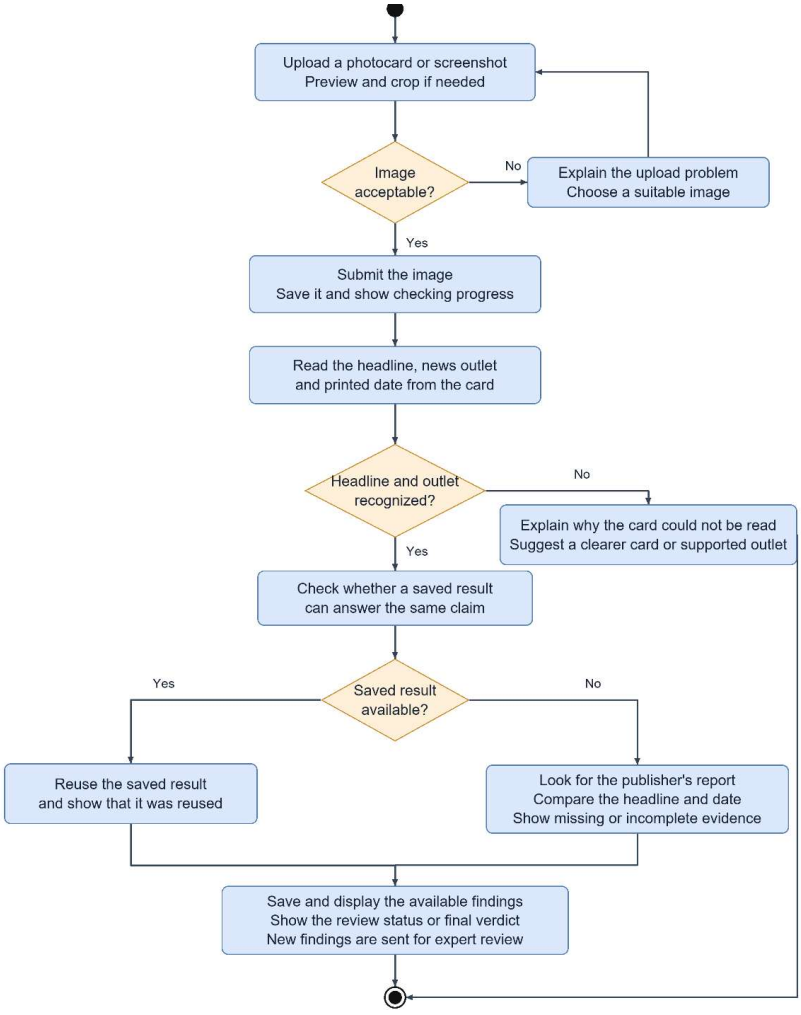


Figure 2.3: Activity Diagram — Photo Card Verification

The user uploads a photo card. The system reads its headline and any visible outlet and date, then checks the extracted claim against news reports. A recognised outlet defines the search scope. If the outlet is missing or unrecognised, the current system can search active registered sources. An unreadable headline or a failed image-reading service prevents the check and produces an explanation.

2.4 Text & Image (Multimodal) Verification

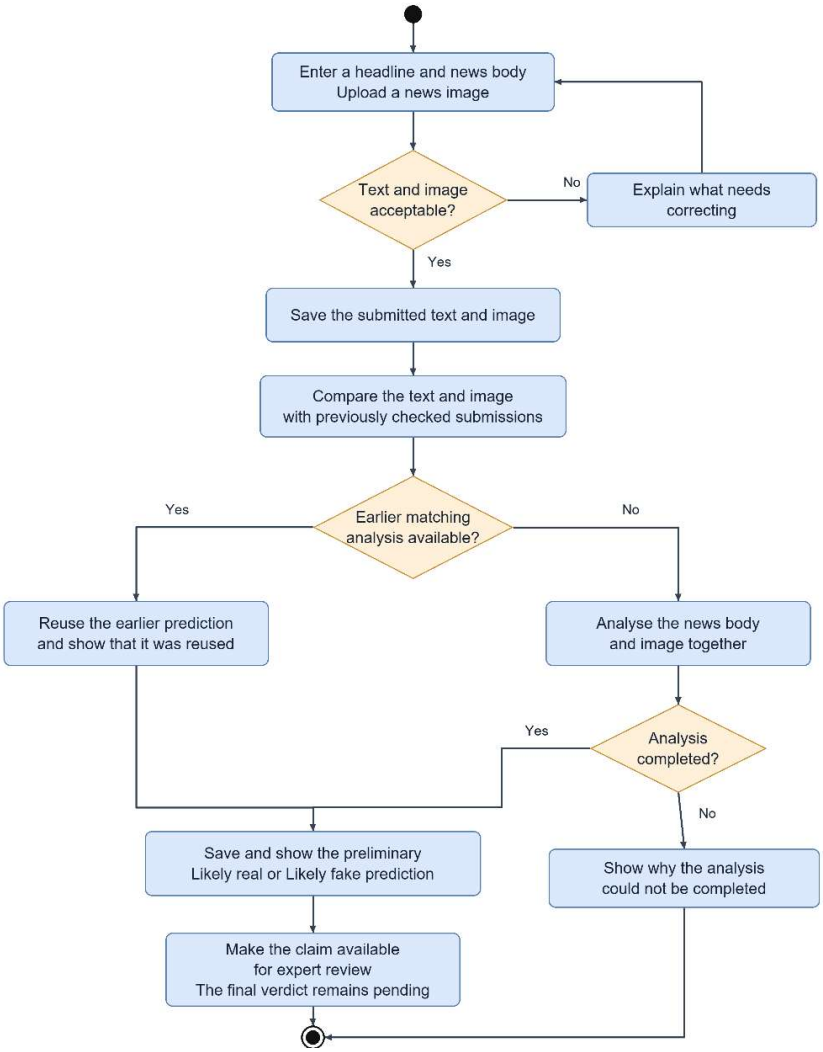


Figure 2.4: Activity Diagram — Text & Image Verification

The user submits a headline, article text and image. The system compares the text and image with earlier analyses. A sufficiently close match can reuse a saved prediction. Otherwise, it analyses the article text and image together and returns Likely Real or Likely Fake. This prediction remains preliminary while the claim awaits expert review.

2.5 Expert Review and Voting

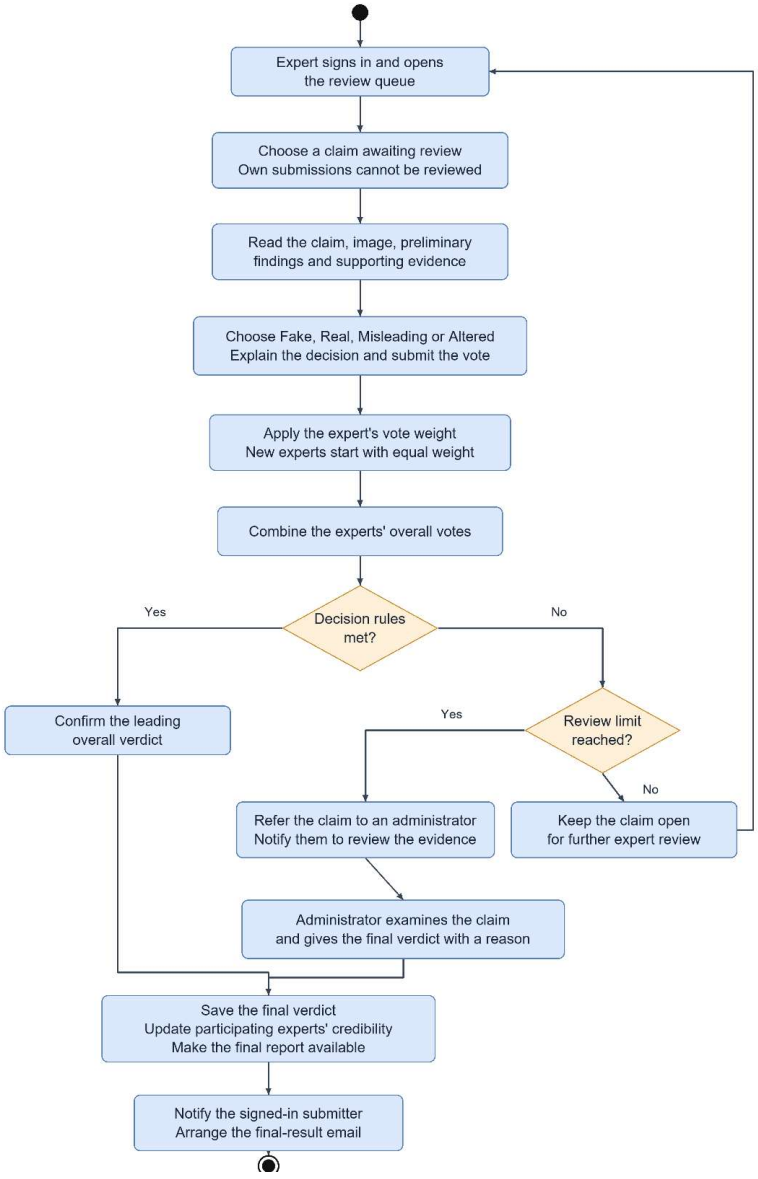


Figure 2.5: Activity Diagram — Expert Review and Voting

An expert selects a claim, reads the evidence and submits a verdict with a written explanation. Experts cannot review their own submissions. The system combines the weighted votes and checks the decision rules. A claim becomes final when those rules are met or goes to an administrator if a review limit is reached first. The final decision is saved and result delivery is scheduled.

2.6 Browser Extension

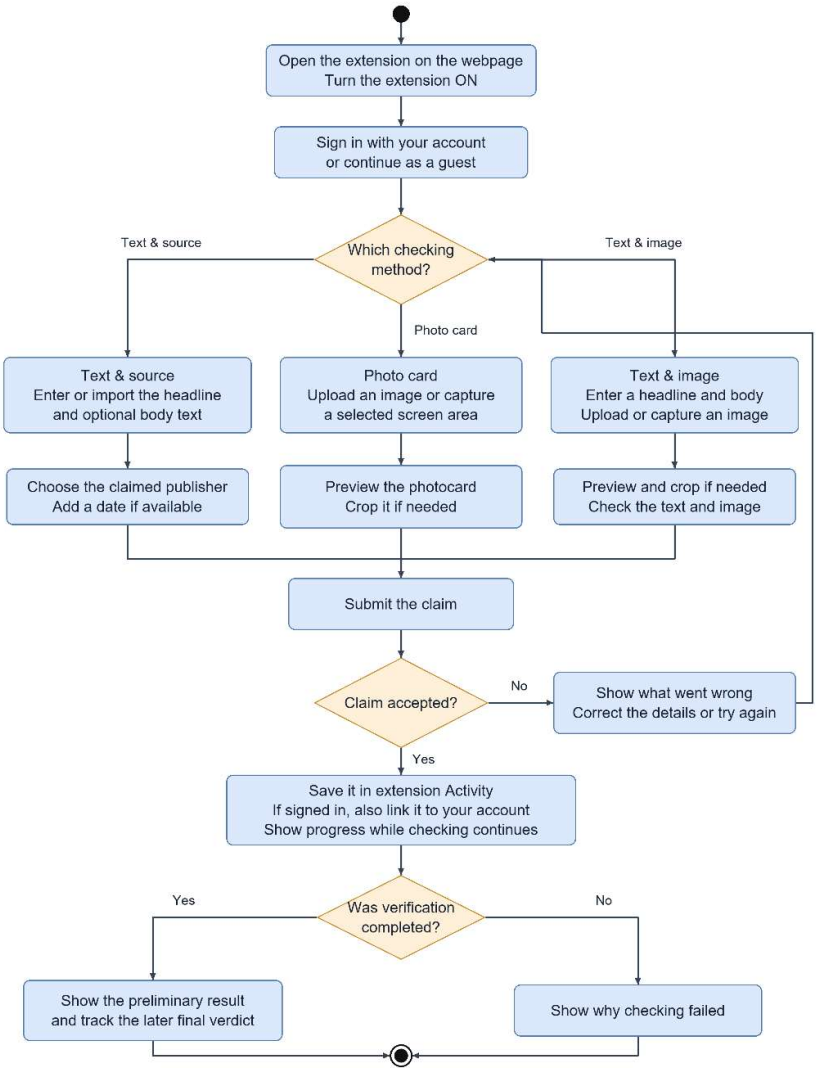


Figure 2.6: Activity Diagram — Browser Extension

The extension lets users submit and follow claims beside the page they are reading. They can type or import text, upload an image or capture and crop a selected area. Activity shows the progress, available findings and final decision. Failed checks show a reason.

2.7 Fact Explorer

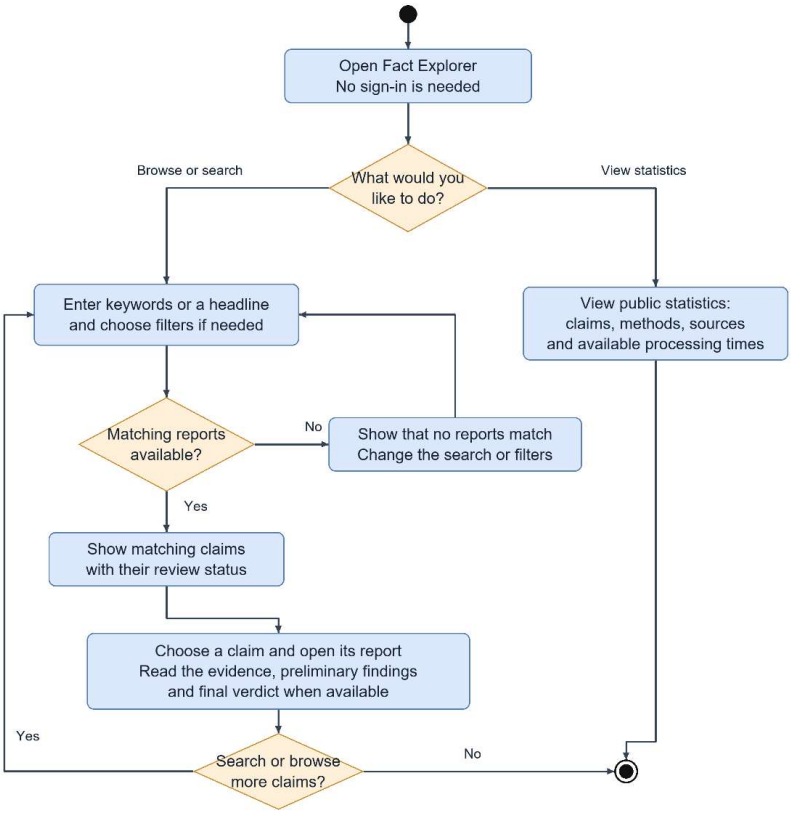


Figure 2.7: Activity Diagram — Fact Explorer

Fact Explorer is available without sign-in. Users can search claims, apply filters and view archive statistics. Each available report shows its evidence and review status, including the final verdict when one has been published.

3. Data-Based Modeling and Dataset Description

3.1 Entity-Relationship Diagram

Figure 3.1 presents the main data entities and their relationships. A submission connects the claim to its evidence, automated findings and reviews. The diagram distinguishes user roles at a conceptual level; the database stores account roles in one user table and keeps additional expert information in a separate profile.

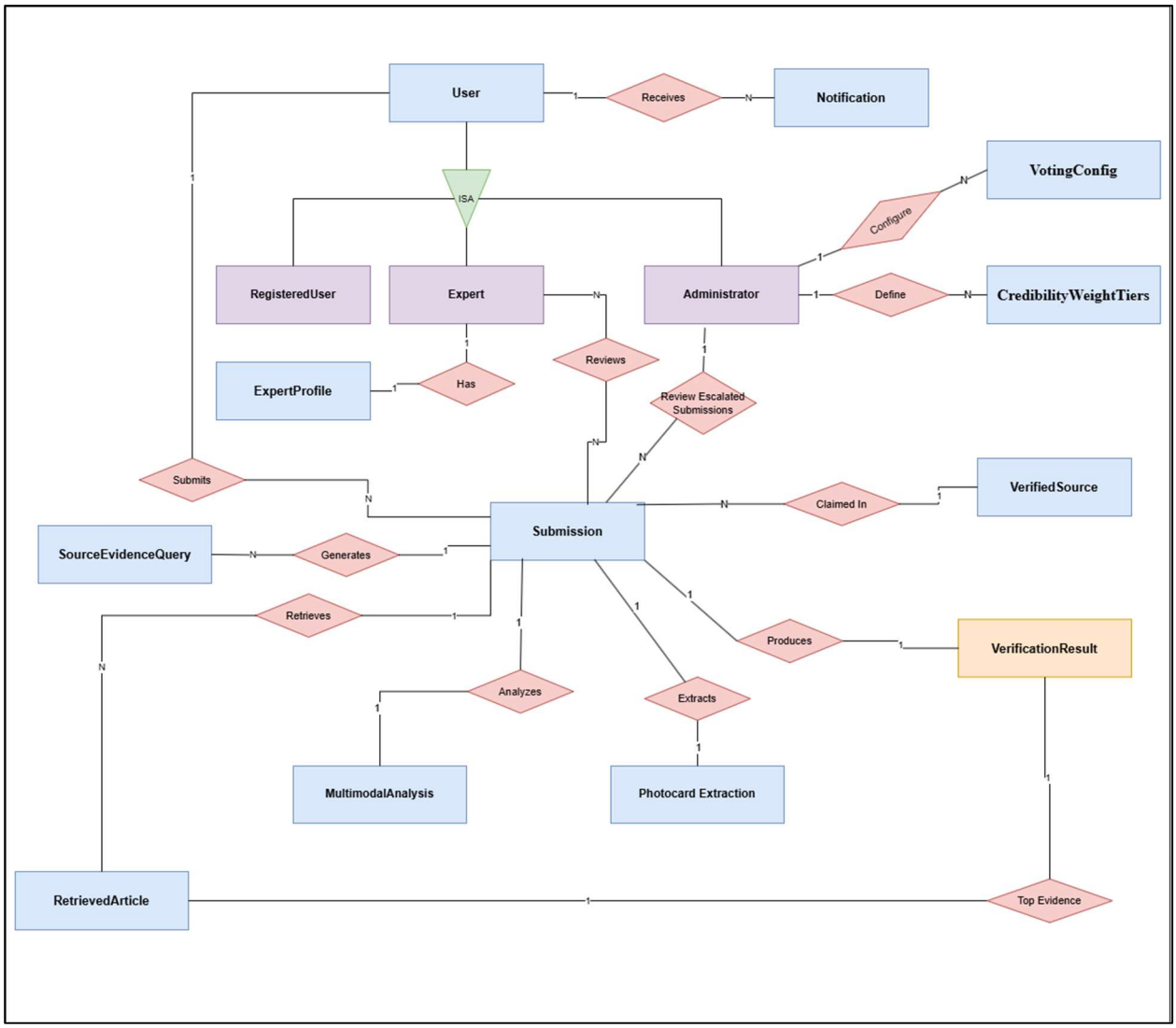


Figure 3.1: Entity-Relationship Diagram — BanglaFactGuard

3.2 Main Entities and Relationships

The main relationships are:

- One account can have many submissions and notifications. Guest submissions do not require an account.

- An expert account can have one expert profile. Each expert can review several submissions and each submission can receive reviews from several experts.
- Administrators manage voting rules and credibility tiers and provide final decisions for escalated submissions.
- A submission can refer to one claimed news source. This relationship is optional for text and image checks and for searches across registered sources.
- A submission can have several search queries and retrieved articles as its evidence record.
- Depending on the verification method, a submission can have a photo card extraction, a multimodal analysis and a verification result. A source-based result can identify a selected evidence article when one is available.

Table 3.1 summarises the main entities and their stored information. Supporting records handle sessions, background jobs and result delivery, as discussed in Section 6.3.

Table 3.1: Main Entities and Their Key Attributes

Entity	Purpose	Main information
User account	Identity and access details	Name, email, password, role and account status
Expert profile	Expertise and review record	Expertise area, completed reviews, agreement with final decisions and credibility score
News source	Registered news publisher	Publisher name, aliases, website, search and extraction settings and active status
Submission	Claim and processing record	Method, headline, article text, claimed outlet and date, submitter, status and failure reason
Evidence article	Retrieved news report	Article address, title, text, publication date and relevance score
Evidence search	Record of an article search	Search text, query type, search provider and number of results
Verification result	Findings and reviewed decision	Source, headline and date findings, comparison details, selected article and final verdict
Photo card extraction	Information read from a card	Image reference, extracted content, reading status, attempts and failure details
Text and image analysis	Preliminary model assessment	Prediction, class confidence scores, text and image representations and model version
Expert review	Reviewer decision and explanation	Reviewer, submission, verdict, justification, vote weight and administrator decision indicator
Credibility Weight Tiers	Weight of experts vote	Min accuracy, max accuracy, multiplier, label, isactive
Voting rules	Decision settings	Minimum voters, required vote weight, lead margin, review limits
Notification	Personal result update	Recipient, title, message, type, result link and read status

3.3 Dataset Description

The text and image model uses the MultiBanFakeDetect dataset. Each item combines news text with an image. The project uses two labels, Real and Fake, to train and evaluate the classifier.

The training notebook contains 7,680 training items, 960 validation items and 960 test items, giving an 80/10/10 split. The training set contains 3,840 items per class and the test set contains 480 items per class. Table 3.2 summarises the dataset sizes recorded in the notebook.

Table 3.2: Dataset Split Summary (Text–Image Pairs)

Subset	Instances	Share of dataset
Training	7,680	80%
Validation	960	10%
Testing	960	10%
Total	9,600	100%

The source and photo card methods use retrieved news articles and pretrained language models. Their evidence depends on the active source list and the reports available when a check runs. These methods were not evaluated using the multimodal test set.

4. Class-Based Modeling

Class-based modelling groups the main responsibilities of the system. Figure 4.1 shows the analysis classes and how they work together. Table 4.1 summarises their responsibilities and collaborators. Chapter 6 describes their implementation as service components.

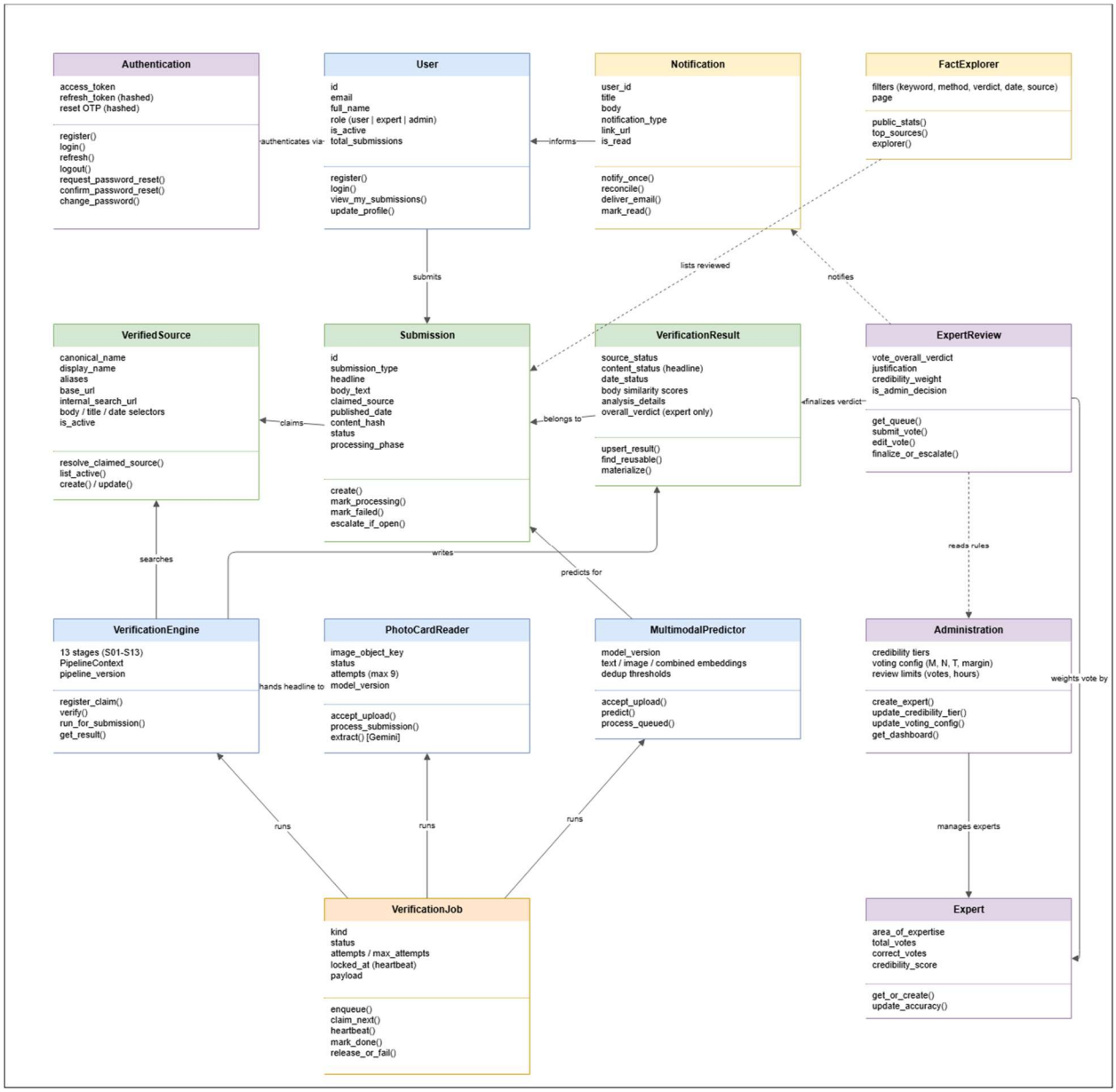


Figure 4.1: Analysis Class Diagram (CRC View)

Table 4.1: Class Responsibilities and Collaborators

Class	Responsibilities	Collaborators
User	Creates an account, signs in and manages personal submissions and settings	Authentication, Submission, Notification
Authentication	Controls sign-in, session renewal, sign-out and password recovery	User
Submission	Stores the claim and tracks its progress and review status	News source, Verification result, Expert review
News source	Identifies registered publishers and their search settings	Submission, Verification engine
Verification engine	Retrieves evidence and prepares source, headline and date findings	Submission, News source, Verification result
Photo card reader	Reads a claim from an image and passes it to the verification engine	Verification engine, Submission
Text and image predictor	Produces preliminary predictions and identifies reusable analyses	Submission
Verification job	Tracks queued work, progress, completion and recovery	Verification engine, Photo card reader, Text and image predictor
Verification result	Stores findings and supports reuse of compatible results	Submission, Expert review
Expert review	Records votes and supports final decisions or escalation	Expert, Administration, Notification
Expert	Maintains an expertise profile and a record of completed reviews	Expert review, Administration
Administration	Manages experts, tiers, voting rules and the dashboard	Expert, Expert review
Notification	Records personal updates and tracks final-result email delivery	User, Expert review
Fact Explorer	Provides access to available claims, filters and archive statistics	Submission

5. AI Engineering Design

AI supports two main tasks: assessing text and images together and comparing a claim with retrieved news reports. This chapter explains the models used for those tasks and the limits of their output. Photo card reading supplies the claim text for the source check; its processing is described in Section 8.3.3.

5.1 Multimodal Verification Model

5.1.1 Model Architecture

Figure 5.1 shows how the model combines text and image features. BanglaBERT represents the article text, while EfficientNet-B4 represents the image. A classifier combines these representations to produce scores for Real and Fake.

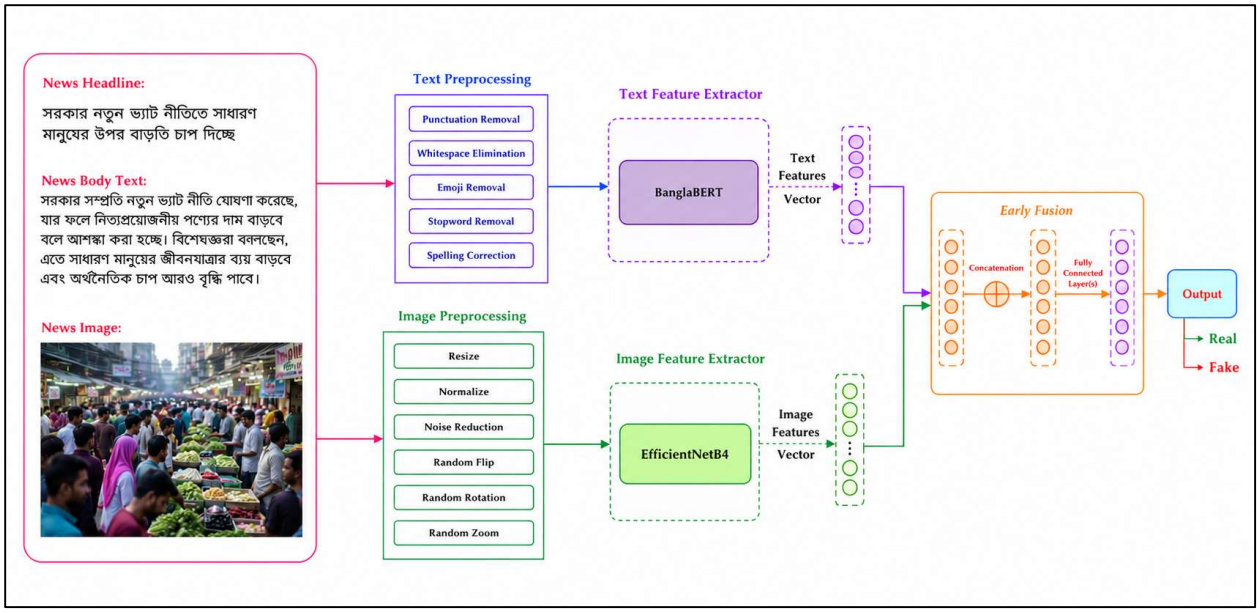


Figure 5.1: Multimodal Verification Architecture — Text and Image Fusion

The model combines 768 text features and 1,792 image features before classification. It uses up to 128 body-text tokens and resizes images to 380 × 380 pixels. The headline is retained for display. Table 5.1 gives the model settings.

Table 5.1: Multimodal Model Configuration

Item	Value
Text encoder	BanglaBERT; 768 text features
Image encoder	EfficientNet-B4; 1,792 image features
Fusion and classifier	2,560 combined features; hidden layers of 1,024 and 512 units; two output classes
Training	35 epochs, learning rate 2e-4, weight decay 0.01, batch size 6, label smoothing 0.1, 3 warm-up epochs
Output	Two class scores, displayed as Likely Real or Likely Fake

5.1.2 Recorded Evaluation

Table 5.2 reports the evaluation saved in the project training notebook. The model achieved 80% accuracy on 960 test items. This result applies to the recorded dataset and evaluation setup; accuracy on other Bangla news requires separate testing.

Table 5.2: Recorded Test Results of the Multimodal Model (960 test items)

Class	Precision	Recall	F1-score	Support
Real (0)	0.79	0.81	0.80	480
Fake (1)	0.81	0.79	0.80	480
Weighted average	0.80	0.80	0.80	960
Accuracy			0.80	960

5.1.3 Duplicate Detection

Before a new prediction, the system compares the text and image representations with recent saved analyses. Reuse requires a close match in the text, the image and their combined representation. Checking all three helps distinguish the same article paired with a different image. The result page identifies a reused assessment.

5.2 Source-Based Verification Pipeline

The 13 stages in Figure 5.2 prepare the claim, retrieve relevant articles and produce separate comparisons. Searches remain within the permitted news sources. The report distinguishes incomplete processing from a completed search with no relevant result.

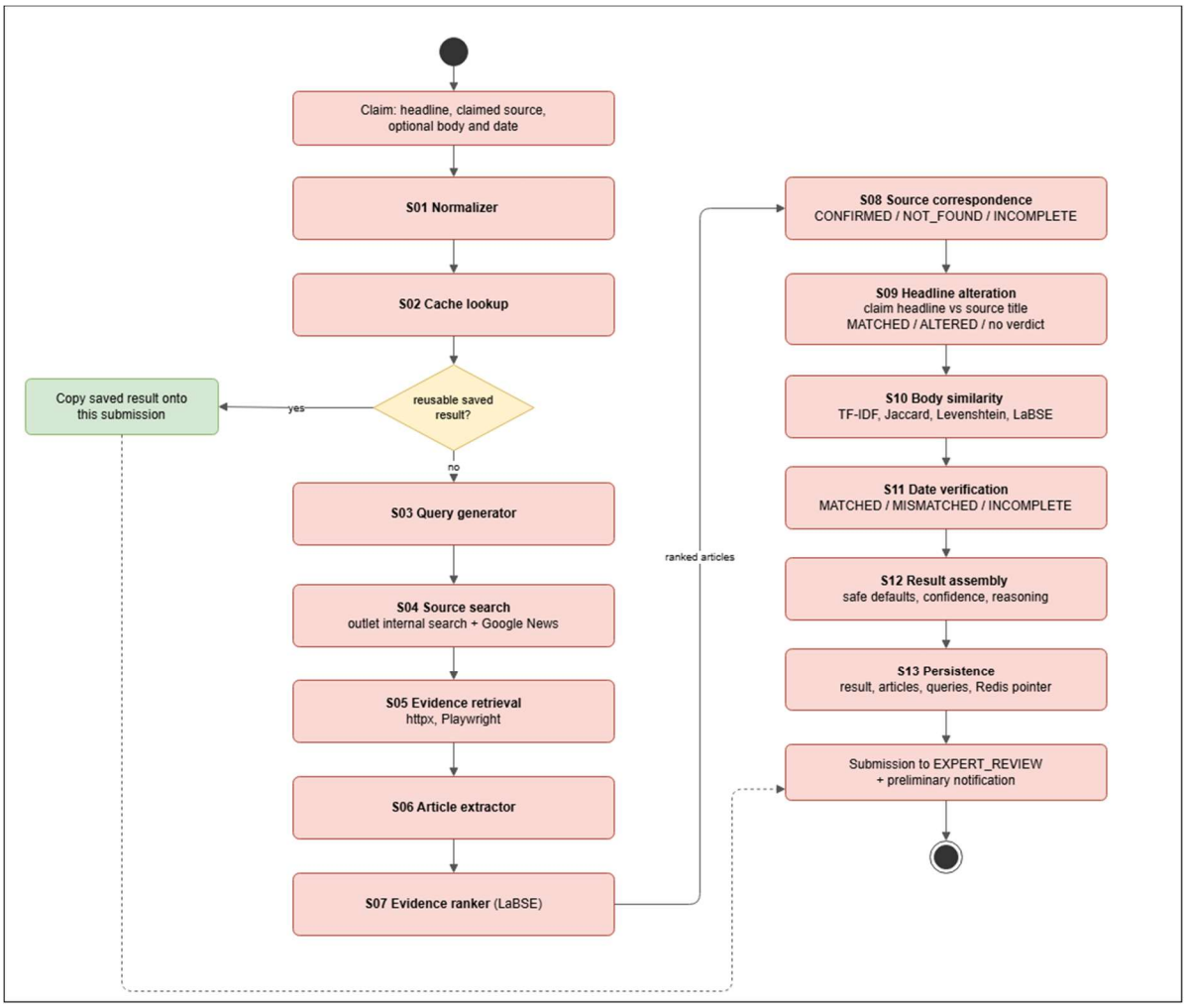


Figure 5.2: Source-Based Verification Pipeline — Stages S01 to S13

Table 5.3: Source-Based Verification Pipeline — Stage Reference

Stage	What it does
S01 Normalizer	Prepares the text and identifies the search scope.
S02 Cache lookup	Checks for a compatible completed result that can be reused.
S03 Query generator	Creates search phrases from the headline, keywords and date.
S04 Source search	Searches permitted publishers and rejects off-source results.
S05 Evidence retrieval	Loads article pages, using browser rendering when needed.
S06 Article extractor	Reads the article title, body and publication date from each page.
S07 Evidence ranker	Ranks articles by their relevance to the claim.
S08 Source correspondence	Selects relevant evidence and records the source finding.
S09 Headline alteration	Compares the headline with the selected source title.

Stage	What it does
S10 Body similarity	Measures body-text similarity when article text is supplied.
S11 Date verification	Compares available claimed and source publication dates.
S12 Result assembly	Prepares the preliminary findings and explanations.
S13 Persistence	Saves the result and schedules any personal notification.

Headline comparison considers changes such as numbers, dates, negation and named people or organisations. It uses language models alongside these checks to assess whether the meaning is preserved. When the evidence is insufficient, the report leaves the comparison undecided and explains why. Body similarity scores describe overlap between texts; they do not determine whether a claim is true.

5.3 Model Selection

Table 5.4 identifies each model and its role. These are the components selected for the implementation, not the outcome of a comparative model benchmark. The recorded evaluation covers the combined text and image classifier.

Table 5.4: AI Model Selection Summary

Model	Used in	Role in the system
BanglaBERT	Text & image (text encoder)	Provides a representation of Bangla article text for classification.
EfficientNet-B4	Text & image (image encoder)	Extracts visual features from the submitted image.
LaBSE (sentence-transformers)	S07–S10	Represents sentence meaning for article ranking and text comparison.
mMARCO MiniLM cross-encoder	S07 re-ranking	Refines the order of shortlisted claim–article pairs.
mDeBERTa-v3 NLI (MNLI/XNLI)	S09	Assesses support or contradiction between the claim headline and source title.
BanglaTag NER	S09 entity checks	Identifies people, places and organisations for headline comparison.

5.4 Responsible Use of AI

- Automated findings are presented as preliminary. A final verdict is published only after the expert voting rules are met or an administrator decides an escalated claim.
- A missing source report does not establish that a claim is false. An incomplete search and an unavailable model score are shown as limitations of the check.
- The source-checking models and thresholds need evaluation on labelled Bangla claims. Tests with simulated responses check software behaviour, not model accuracy. Expert review remains part of the decision process.

6. Component-Level Design

This chapter explains how the interfaces, verification services and data stores work together. It also describes the submission lifecycle, voting rules and local deployment.

6.1 Component Diagram

The website and Chrome extension use the same backend service. The backend handles accounts, verification, expert review and administration. Background workers process accepted jobs and deliver results. PostgreSQL stores application records, Redis caches reusable data and MinIO stores uploaded images.

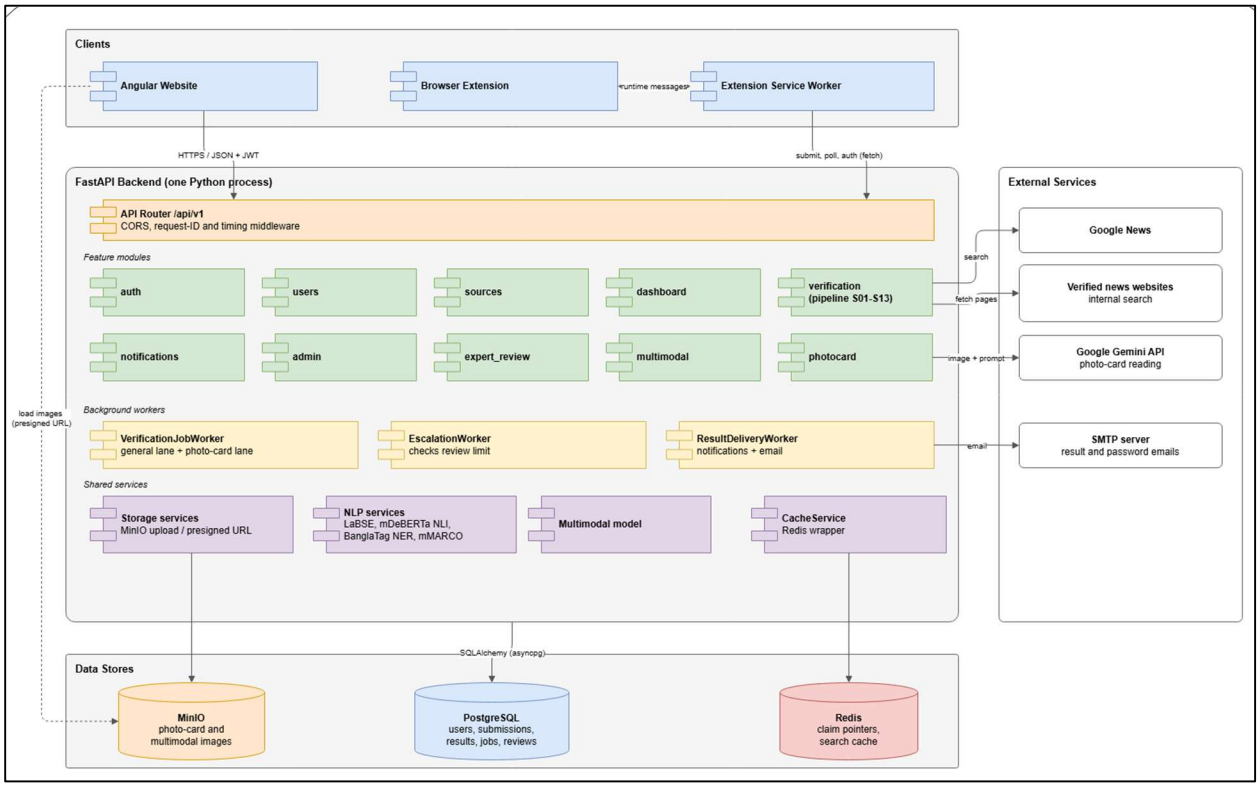


Figure 6.1: Component Diagram of BanglaFactGuard

Table 6.1: Main Components and Their Responsibilities

Component	Responsibility
Angular website	Public pages, claim forms, result reports, history, notifications, expert and admin workspaces.
Browser extension	Accepts claims, tracks progress and displays activity and result alerts.
API layer	Receives client requests and directs them to the appropriate service.
Account management	Handles registration, sign-in, password recovery, profiles and personal history.
Source verification	Retrieves evidence and compares the submitted claim with relevant reports.

Component	Responsibility
Photo card processing	Reads a claim from an uploaded card and sends it for source verification.
Text and image analysis	Checks for reusable analyses and produces preliminary predictions.
Expert review	Manages the review queue, votes, final decisions and escalation.
Administration and sources	Manages experts, publishers, credibility tiers and voting rules.
Public archive and notifications	Provides claim search and statistics and records personal updates.
Job processing	Runs saved verification jobs, with separate capacity for photo card processing.
Review monitoring	Checks review limits and escalates unresolved claims.
Result delivery	Sends personal result notifications and final-result emails.
Shared services	Provides image storage, language analysis, model inference and caching.

6.1.1 Information Flow

A Text & source claim moves through the components as follows:

1. The user submits a claim from the website. A signed-in submission is linked to that account.
2. The backend validates the input, determines the search scope and saves the claim with its background job.
3. The website receives a submission reference and checks its status while the result page is open.
4. A background worker takes the queued job and starts verification.
5. The verification service retrieves relevant news articles and compares the available evidence with the claim.
6. The findings are saved, the claim enters expert review and a personal update is created for a signed-in submitter.
7. Experts review the claim. After finalisation, the result is updated and final-result delivery is scheduled.

6.2 Design Classes

Figure 6.2 shows the design classes. Table 6.2 summarises their service responsibilities and main operations.

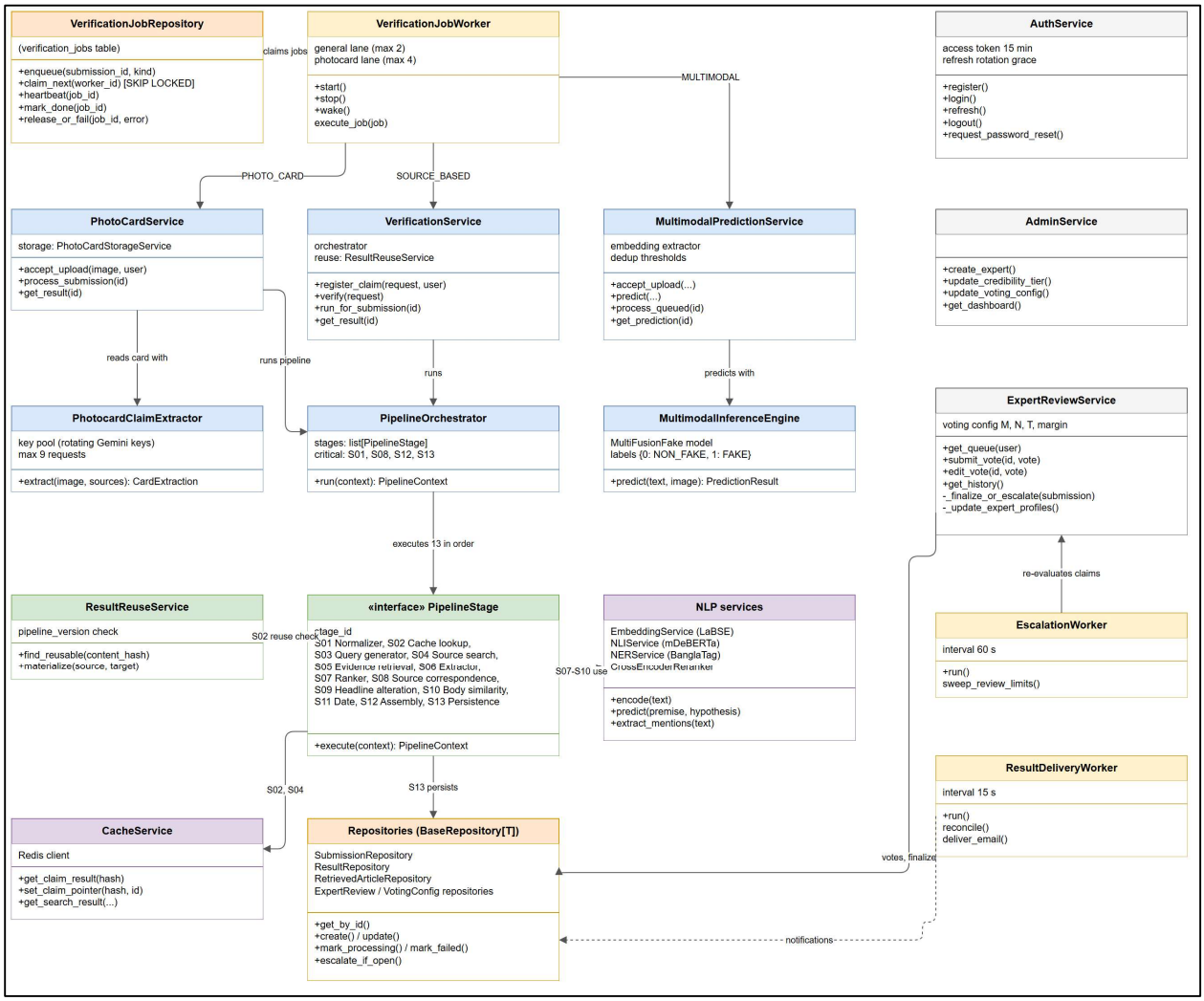


Figure 6.2: Design Class Diagram

Table 6.2: Design Classes and Domain Concepts

Domain concept	Service responsibility	Main actions
Claim checking	Verification service and processing stages	Accept claims, run the evidence check and assemble findings
Photo card	Photo card service and claim reader	Store the image, extract the claim and start verification
Text & image	Prediction service and inference engine	Accept text and images and produce preliminary predictions
Background work	Job queue and worker	Queue work, claim jobs, track progress and recover interrupted processing
Result reuse	Result reuse and cache services	Find compatible results and attach reusable findings to a submission
Expert voting	Review service and review monitor	Record votes, apply decision rules and escalate unresolved claims
Accounts and admin	Authentication and administration services	Manage sessions, expert accounts, sources and voting settings
Notifications	Notification and delivery services	Record personal updates and send final-result emails
Data access	Data repositories	Read and update application records through a common data-access layer

6.3 Persistent Data Sources

Figure 6.3 shows the data stores used by the backend. Table 6.3 distinguishes durable application records from temporary cached data and browser storage.

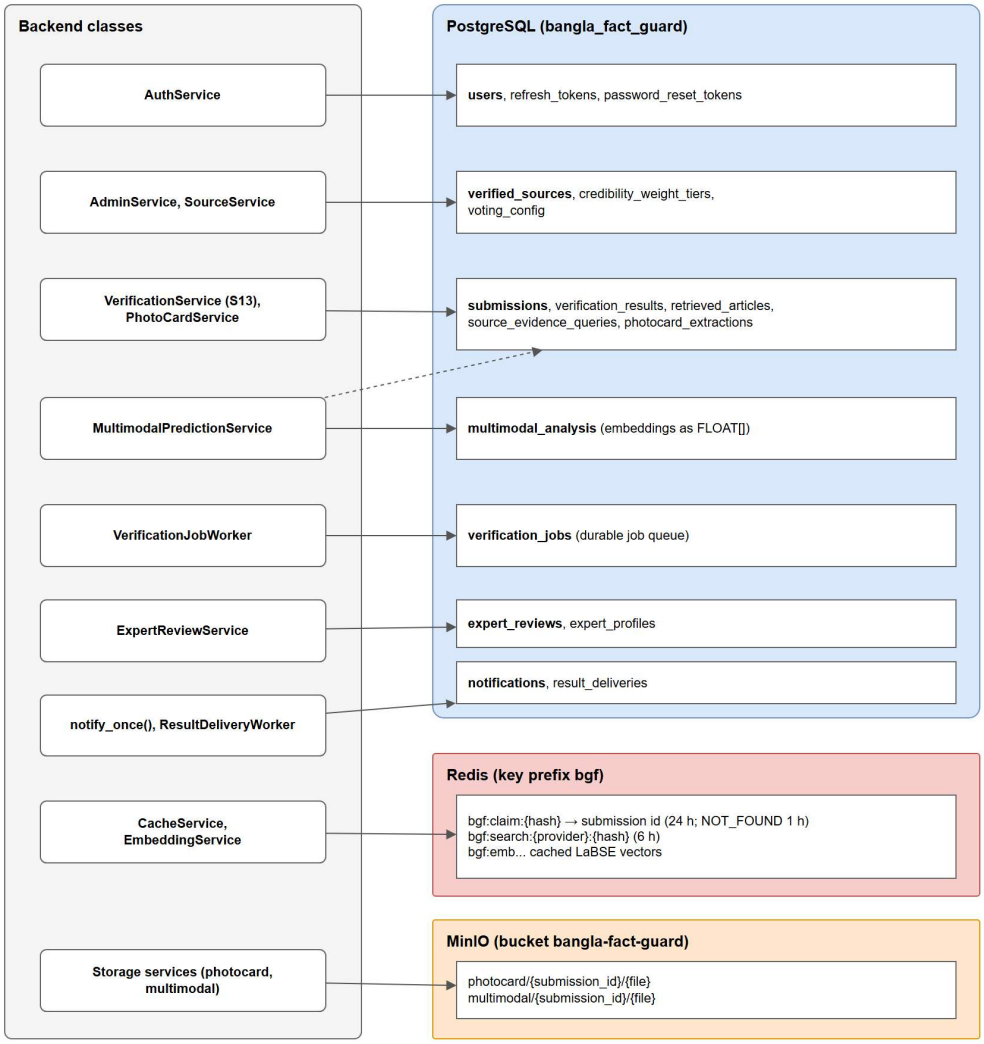


Figure 6.3: Persistent Data Sources and Their Classes

Table 6.3: Persistent Data Sources

Data store	What is stored	Lifetime
PostgreSQL database	Accounts, sources, claims, evidence, results, reviews, notifications, background jobs and delivery records	Retained until removed through database or application maintenance
Redis cache	Search results, text representations and references to reusable verification results	Temporary; entries expire according to the cache settings
MinIO image storage	Uploaded photo cards and news images, accessed through temporary signed links	Objects remain until deleted; access links expire
Extension browser storage	Preferences, sessions, drafts and account-separated activity	Depends on the record type and account or browser cleanup

6.4 Behavioural Design

6.4.1 Submission Lifecycle

Figure 6.4 shows the main submission states. The system also records the current processing step so that users can distinguish queued work from image reading and verification.

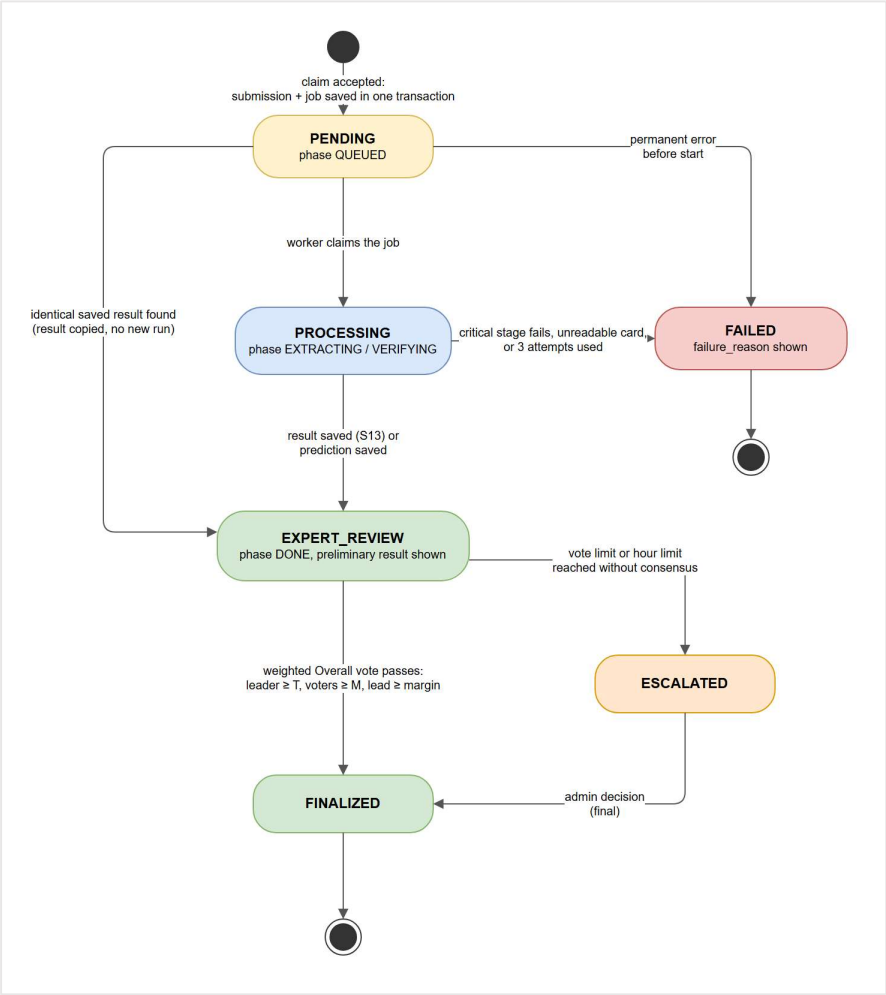


Figure 6.4: State Diagram of a Submission

Table 6.4: Submission States

State	Meaning	Leaves when
PENDING	The claim has been accepted and is waiting for processing	Processing starts, a saved result is reused or the request fails
PROCESSING	The claim is being read or verified	A preliminary result is saved or processing fails
EXPERT_REVIEW	Preliminary findings are available for human review	The voting rules are met or a review limit is reached
ESCALATED	Waiting for an administrator	The administrator submits the final decision
FINALIZED	Final verdict published	End state
FAILED	Processing has stopped and a reason is available	End state

6.4.2 Sequence of a Text & Source Check

Figure 6.5 follows a source-based claim from submission to review. It shows the status checks made by the website, the background verification work and the later delivery of a final decision.

6.4.3 Voting Rules

Each expert vote has a stored weight. New experts use weight 1 until they reach the configured number of completed reviews. Later votes use the credibility tier assigned from their review record. A final decision requires a clear leading verdict, enough voters, sufficient total weight and the required lead over the next verdict. A tie remains unresolved. Claims that reach the vote or time limit go to an administrator for a final decision.

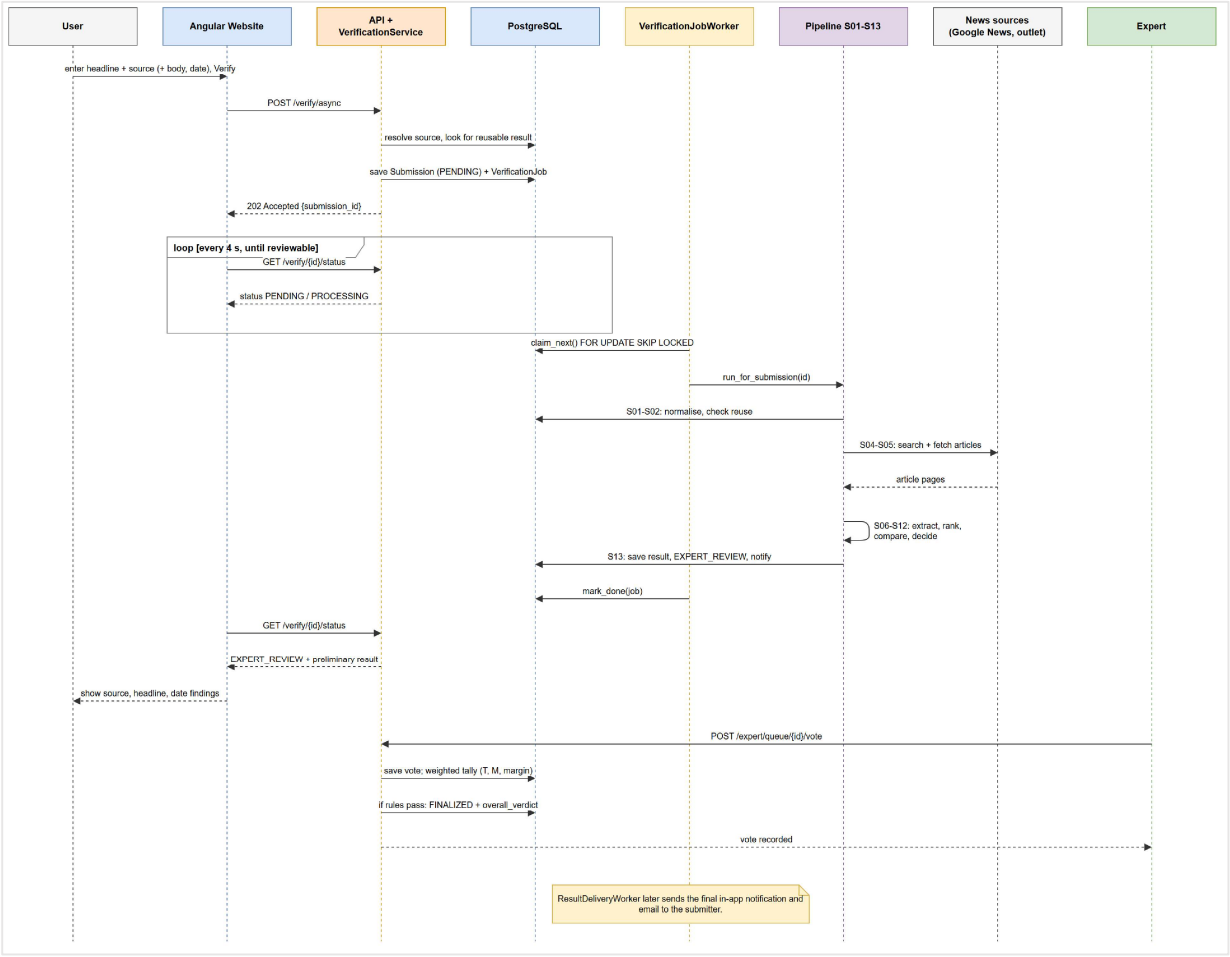


Figure 6.5: Sequence Diagram — Text & Source Claim (rotated to fit the page)

6.5 Deployment Design

Figure 6.6 shows the local development setup used for the report. The browser loads the website and communicates with the backend. PostgreSQL runs on the host, while Docker Compose

provides Redis and MinIO. This setup demonstrates the application locally; it is not evidence of a public production deployment.

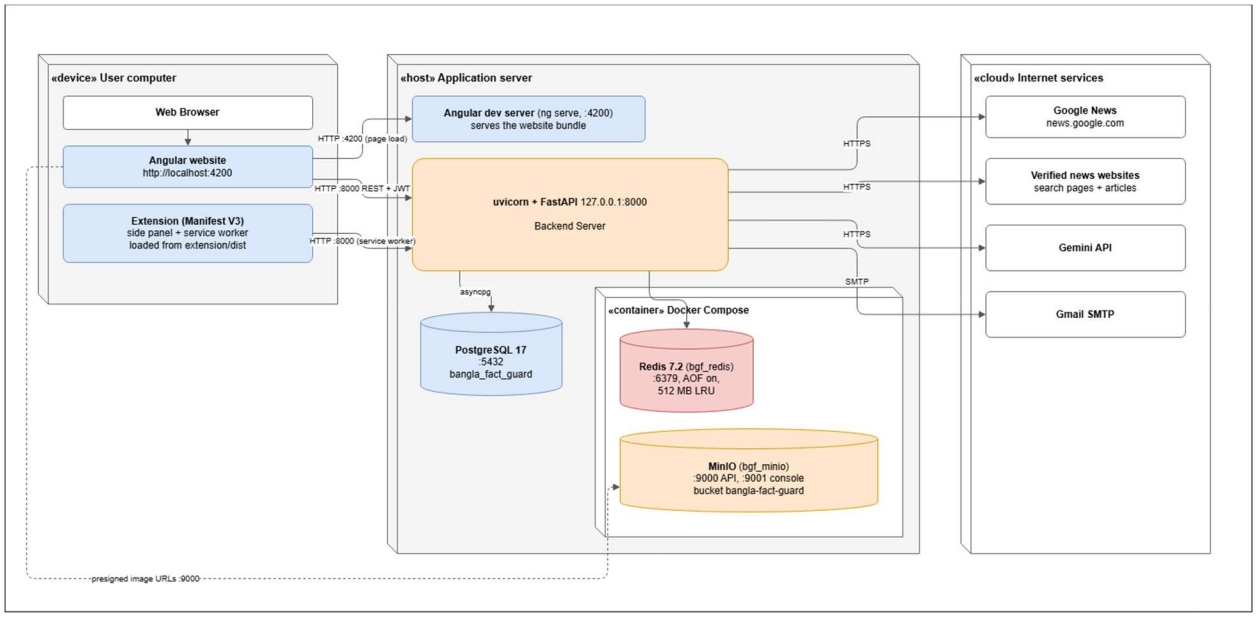


Figure 6.6: Deployment Diagram

Table 6.5: Deployment Nodes

Node	Contents	Notes
User computer	Chrome, the website and the browser extension	Connects to the local application services
Application server	Website development server and backend with background workers	Workers run within the backend application process
PostgreSQL 17	Project database on the host	Installed on the host, not in Compose
Docker Compose	Redis cache and MinIO image storage	Managed by the project Compose configuration
Internet services	News search, registered publishers, image-reading service and email server	Require network access and any necessary credentials

6.6 Design Alternatives Considered

Table 6.6 compares the main design choices with possible alternatives and explains the reasons for the current approach.

Table 6.6: Design Alternatives and Decisions

Decision	Alternative considered	Reason for the choice
Durable job table in PostgreSQL	Separate broker (Celery with Redis)	Keeps accepted jobs with the submission records and supports recovery without a separate task broker.
Separate Source, Headline and Date findings	One automated True/False/Partially True label	Shows what was checked and separates missing evidence from a judgement that a claim is false.
Headline compared with the source title only	Headline compared with the whole article body	Compares equivalent parts of the claim and report and keeps body similarity as a separate measurement.
Gemini reading of photocards	Tesseract OCR with manual correction	Reads the headline, outlet and date together from the card. No comparative OCR accuracy result is reported.
Redis stores only a pointer; the database is authoritative	Cache full results in Redis	Keeps the saved database result as the authoritative record.
In-process workers	Separate worker processes	Keeps the local setup smaller while retaining a persistent job queue.

7. Interface Design

The website supports claim checking, results, personal records and review work. The extension supports submission and tracking while browsing. This chapter describes the main controls, navigation and visible states of both interfaces.

7.1 User Interface Objects and Actions

Table 7.1: User Interface Objects and Actions

Interface	UI object	Actions
Header	Navigation links, bell icon, user menu	Open Verify facts, Fact Explorer, My Submissions, Expert Queue, Administration; open notifications; sign out
Verify facts	Method cards	Choose News story against an image, News story against claimed source or Photocard against claimed source
Verify facts	Claim form	Type or paste headline and text; choose outlet and date; upload image; Verify
Verification result	Report sections	Read claim, final decision, preliminary result, article, headline comparison, body scores and date; open the original source; copy the report link
Fact Explorer	Statistics, tabs, search and filters	Switch Final decisions / Under review; search; filter by decision, type, date and outlet; page through results
My Submissions, Notifications	Lists	View details; mark as read; mark all as read
Expert workspace	Queue, review page, vote form	Search; Review & vote; choose verdict; add optional supplementary findings; write justification; Submit vote
Administration	Dashboard, tables and forms	Create, edit, deactivate experts and reset passwords; register, edit, disable and delete sources; edit tiers and voting rules; decide escalated claims
Extension	ON/OFF switch, Verify, Activity and Account tabs	Use selected text; upload image; select screenshot area; crop; submit; refresh; mark read; view details; sign in/out

7.2 Navigation

Figure 7.1 shows the pages available to each role. All three verification methods open a shared report page, with details suited to the selected method.

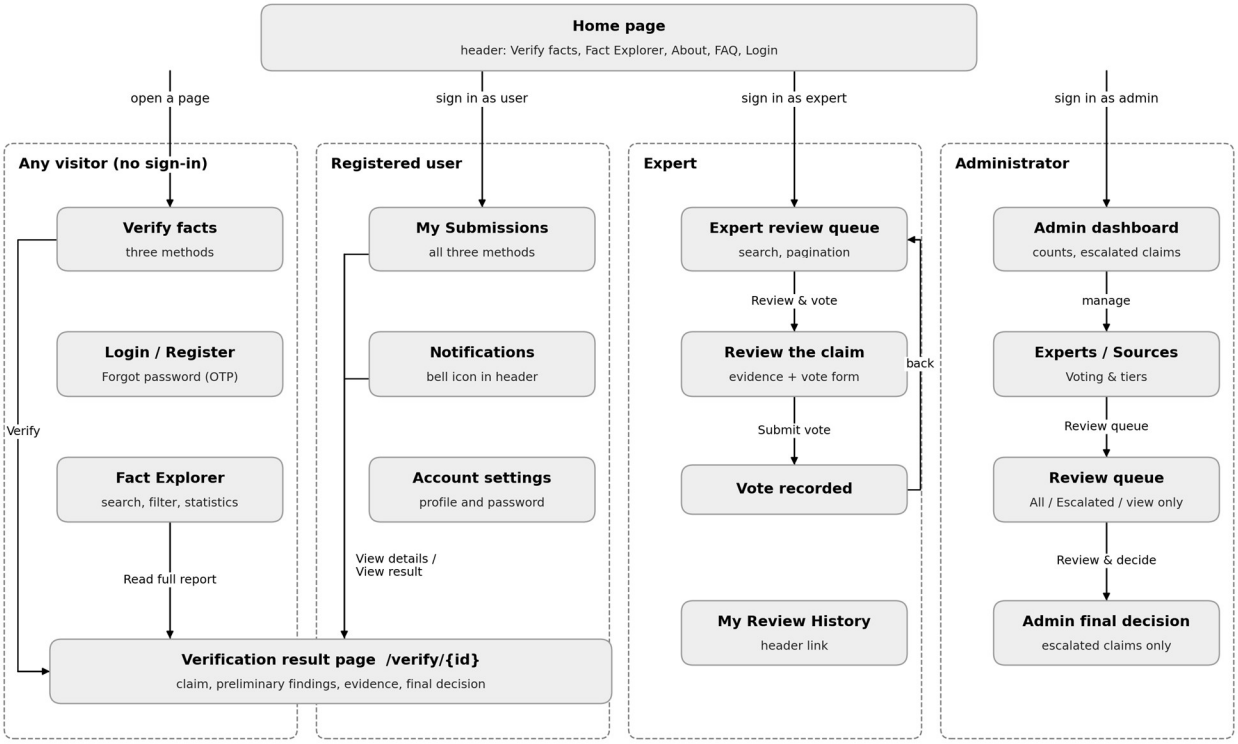


Figure 7.1: Website Navigation by User Role

7.3 UI Events and State Transitions

7.3.1 Verify Facts Page

Figure 7.2 shows the verification page states. Table 7.2 links common user actions and system responses to the state shown on screen.

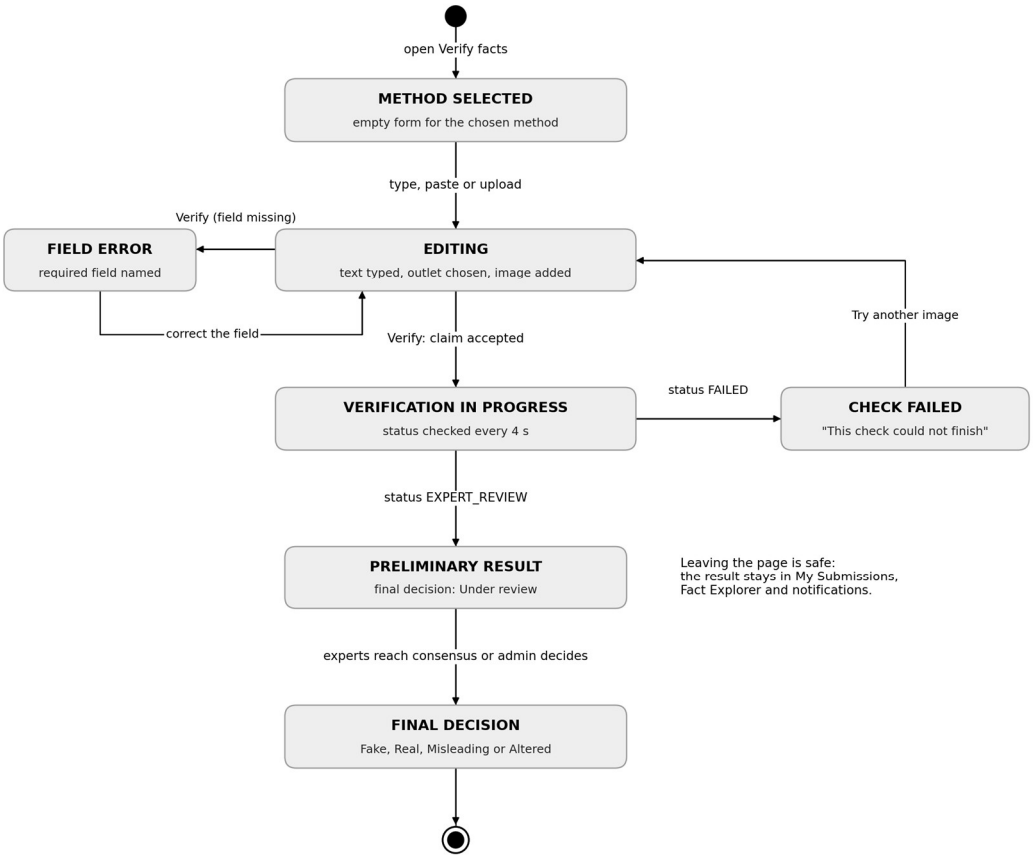


Figure 7.2: State Diagram of the Verify Facts Interface

Table 7.2: Events and State Transitions on the Verify Facts Page

User or system event	From state	To state	What the user sees
Choose a method card	none	METHOD SELECTED	The matching form and a preview panel
Type, paste or upload	METHOD SELECTED	EDITING	Live preview of the claim or image
Click Verify with a required field empty	EDITING	FIELD ERROR	The field is named and the inputs are kept
Click Verify with valid input	EDITING	IN PROGRESS	"Verification in progress" with View progress and My Submissions
Preliminary findings become available	IN PROGRESS	PRELIMINARY RESULT	Findings with Final decision: Under review
Processing fails	IN PROGRESS	CHECK FAILED	"This check could not finish" with the reason
Experts or admin finalize	PRELIMINARY RESULT	FINAL DECISION	Final decision badge and Voting Details

7.3.2 Expert and Administrator Review

The review page lets an expert inspect the claim and submit a verdict. A justification must contain at least 50 characters before submission. Escalated claims appear in the administrator review queue and require an administrator decision.

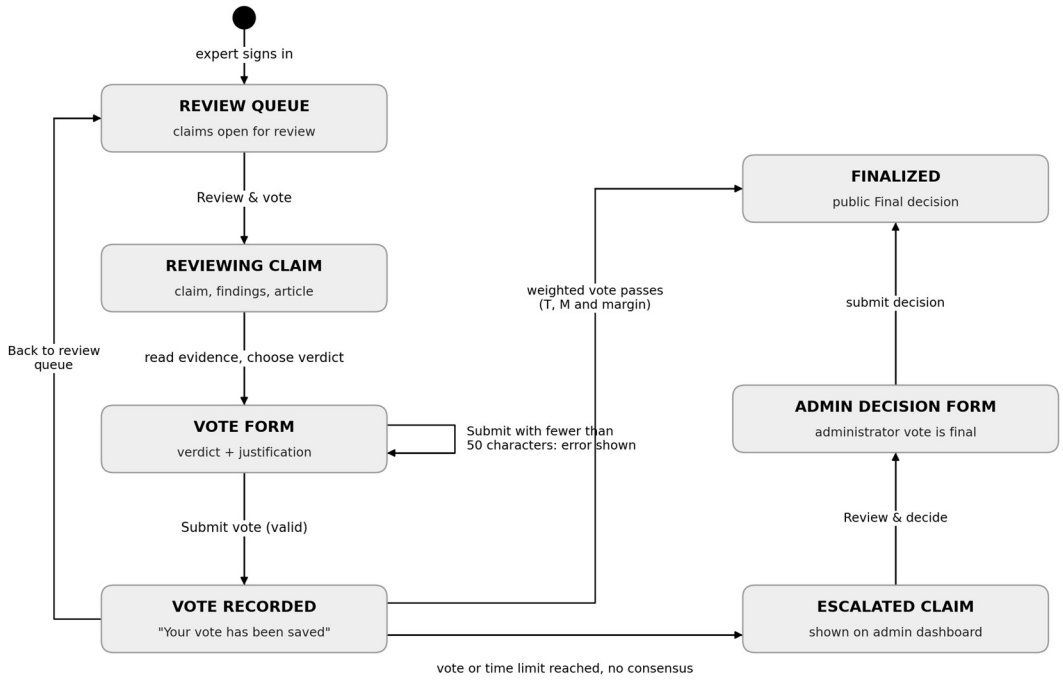


Figure 7.3: State Diagram of the Review Interface

7.3.3 Browser Extension Side Panel

The extension shows separate states for editing, processing and reading results. Switching it off pauses new extension activity and alerts; accepted server jobs continue. The Account tab controls the extension session independently of the claim form.

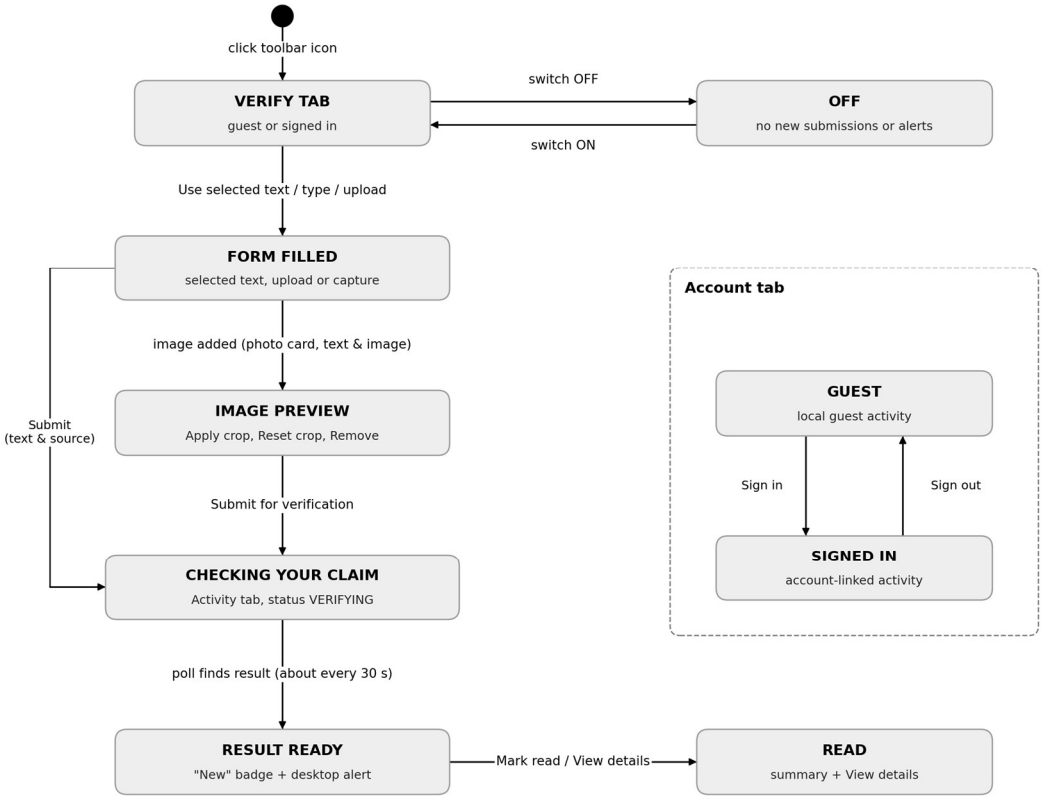


Figure 7.4: State Diagram of the Extension Side Panel

7.4 Interface States as Seen by the User

The screenshots below show the main interface states. They illustrate the forms, progress messages, findings and review controls discussed in this chapter.

Figure 7.5 shows a text and source claim being edited. The preview reflects the entered information. After submission, the page confirms that verification is in progress, as shown in Figure 7.6.

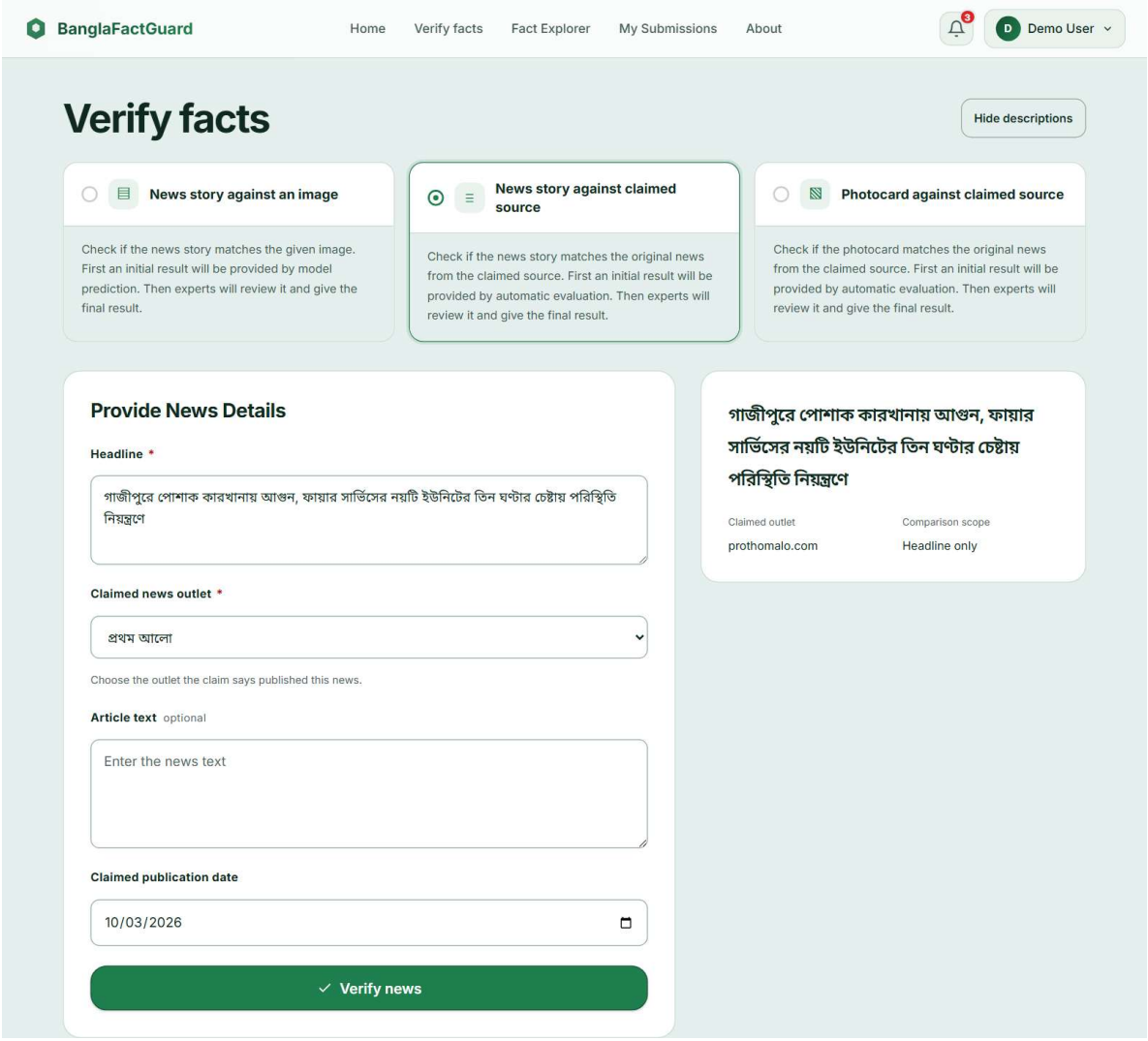


Figure 7.5: EDITING State — Text & Source Claim Form

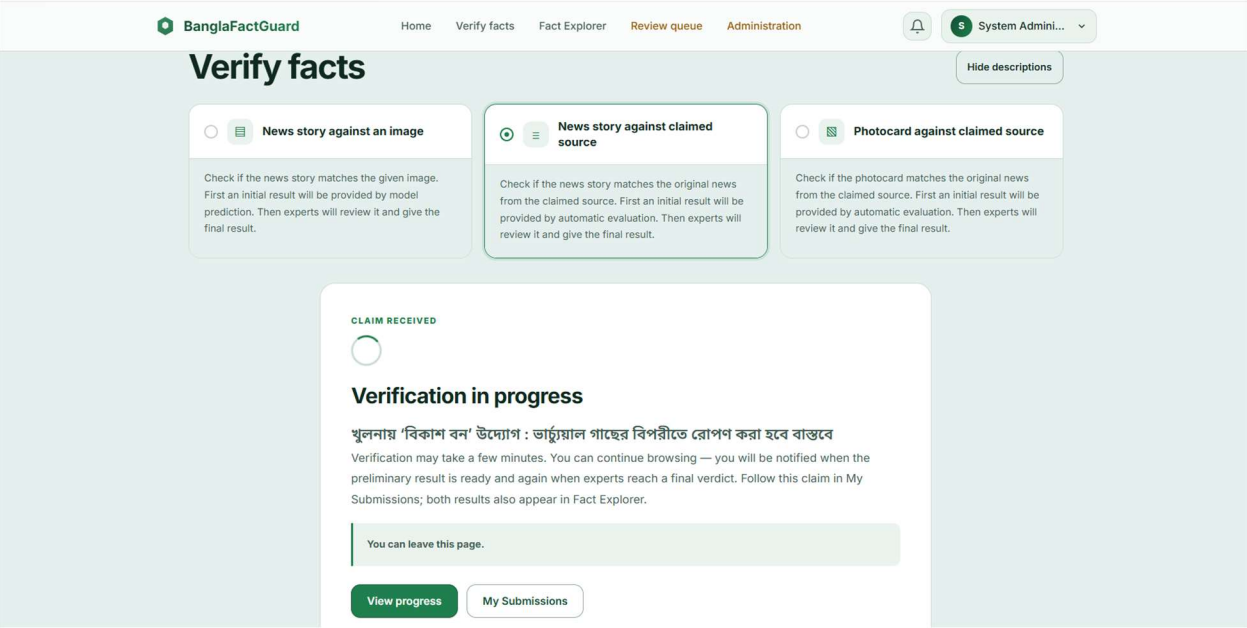


Figure 7.6: IN PROGRESS State — Claim Received

After the automated check, the report displays preliminary findings and shows the final decision as Under review (Figure 7.7). Once review is complete, it displays the verdict and voting details. Figure 7.8 shows a photo card decided by an administrator.

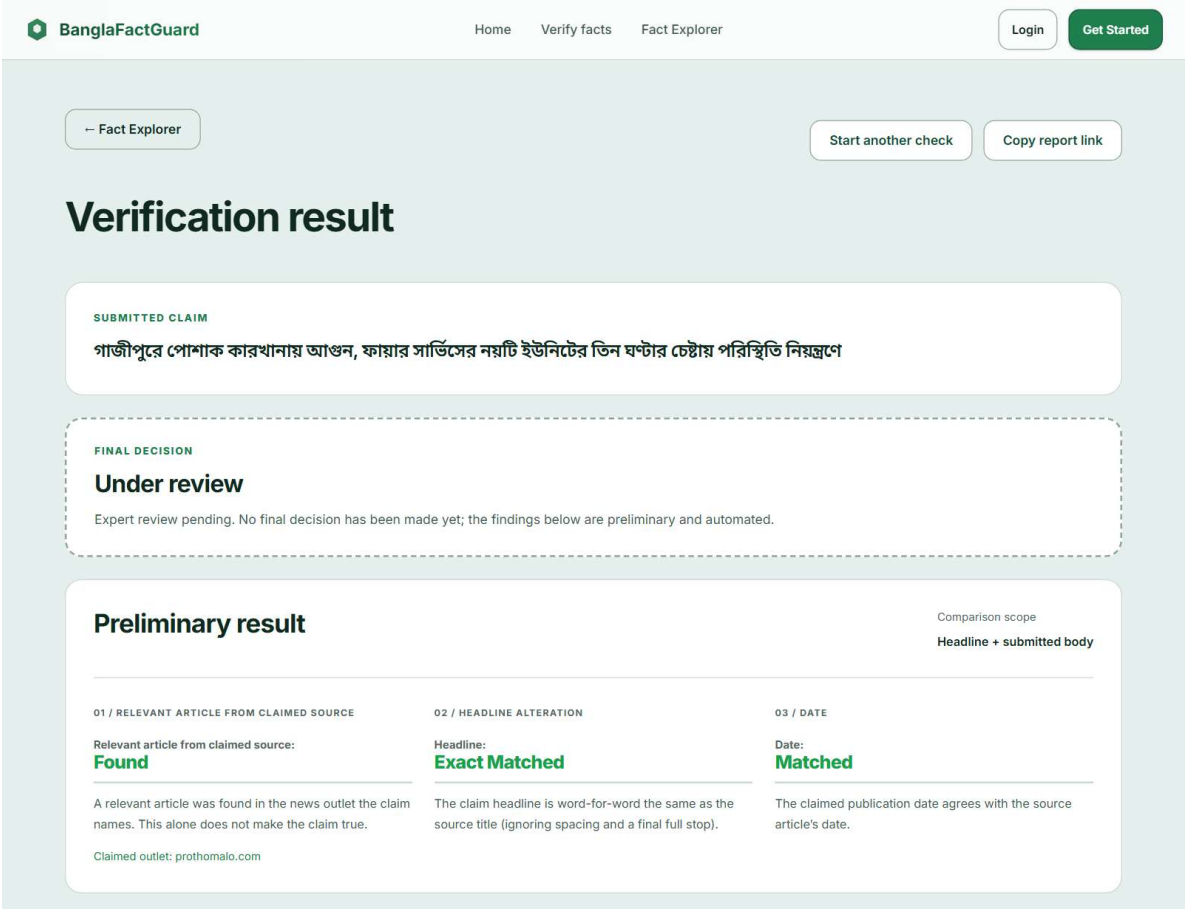


Figure 7.7: PRELIMINARY RESULT State — Final Decision Under Review

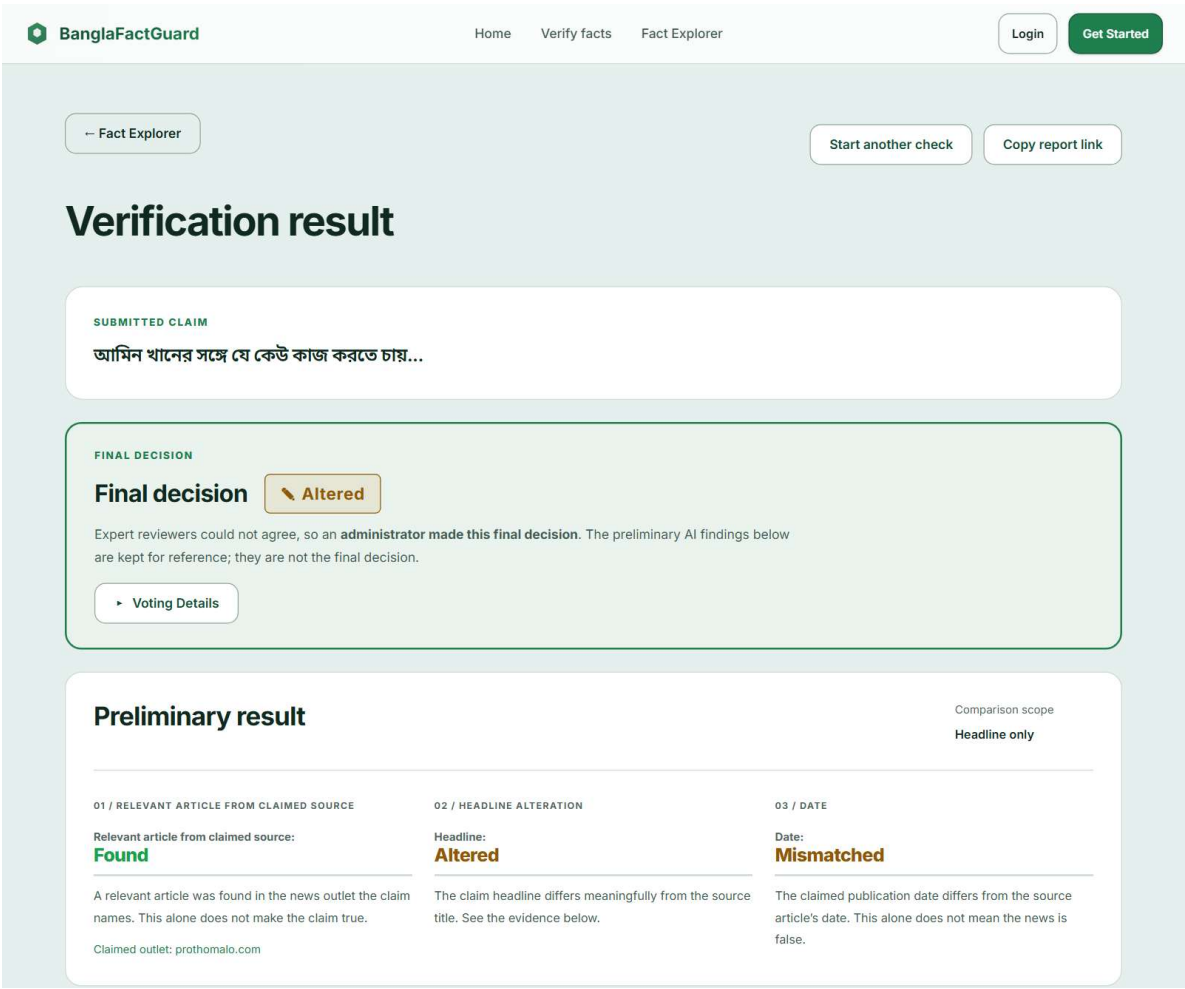


Figure 7.8: FINAL DECISION State — Administrator Decision on a Photo Card

A failed check displays its reason and an available recovery action (Figure 7.9). The extension Activity view also distinguishes claims being processed from results that are ready (Figure 7.10).

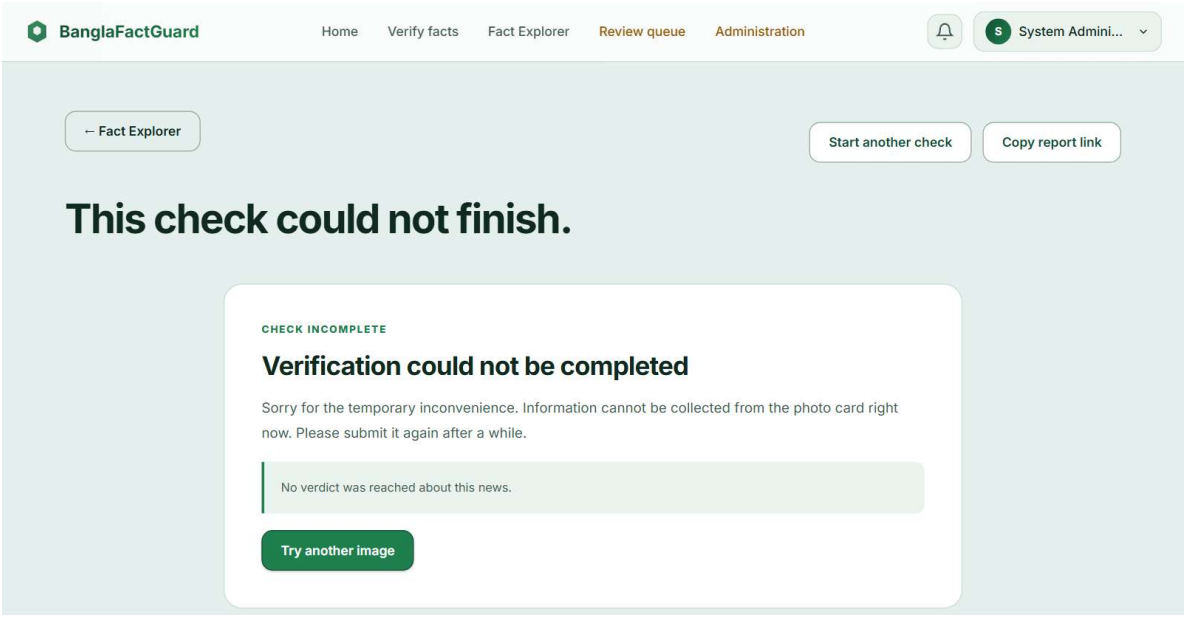


Figure 7.9: CHECK FAILED State — Photo Card Could Not Be Read

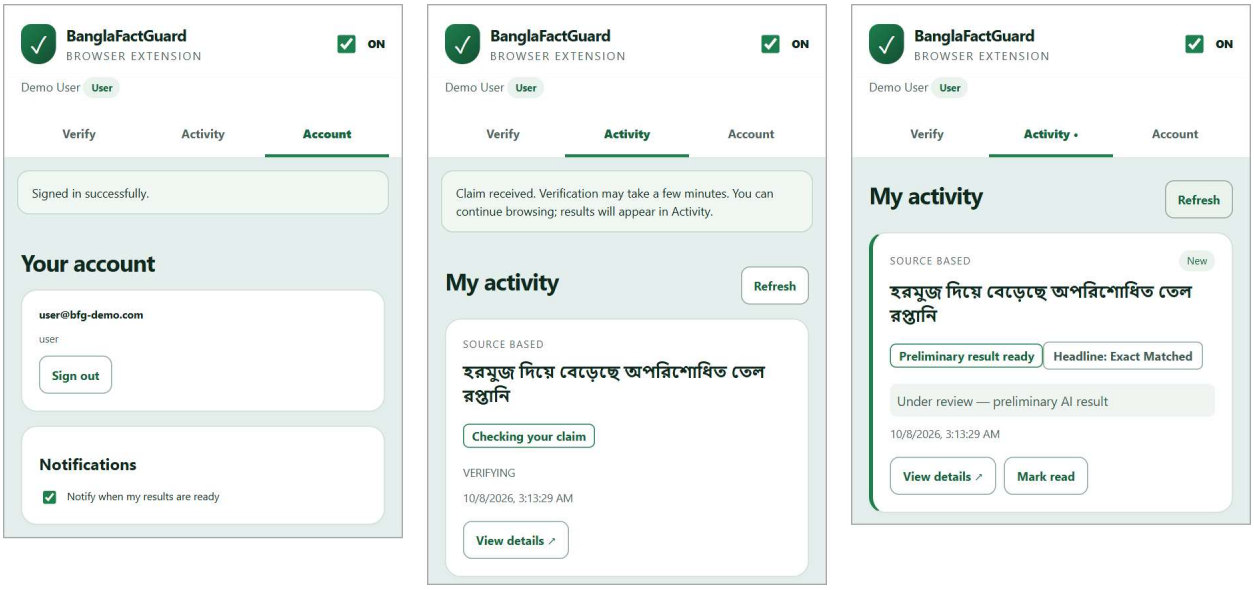


Figure 7.10: Extension States — Signed In, Checking Your Claim and Result Ready

Figure 7.11 shows the expert review form beside the claim and its evidence. The form includes the verdict options and a justification field with a character counter.

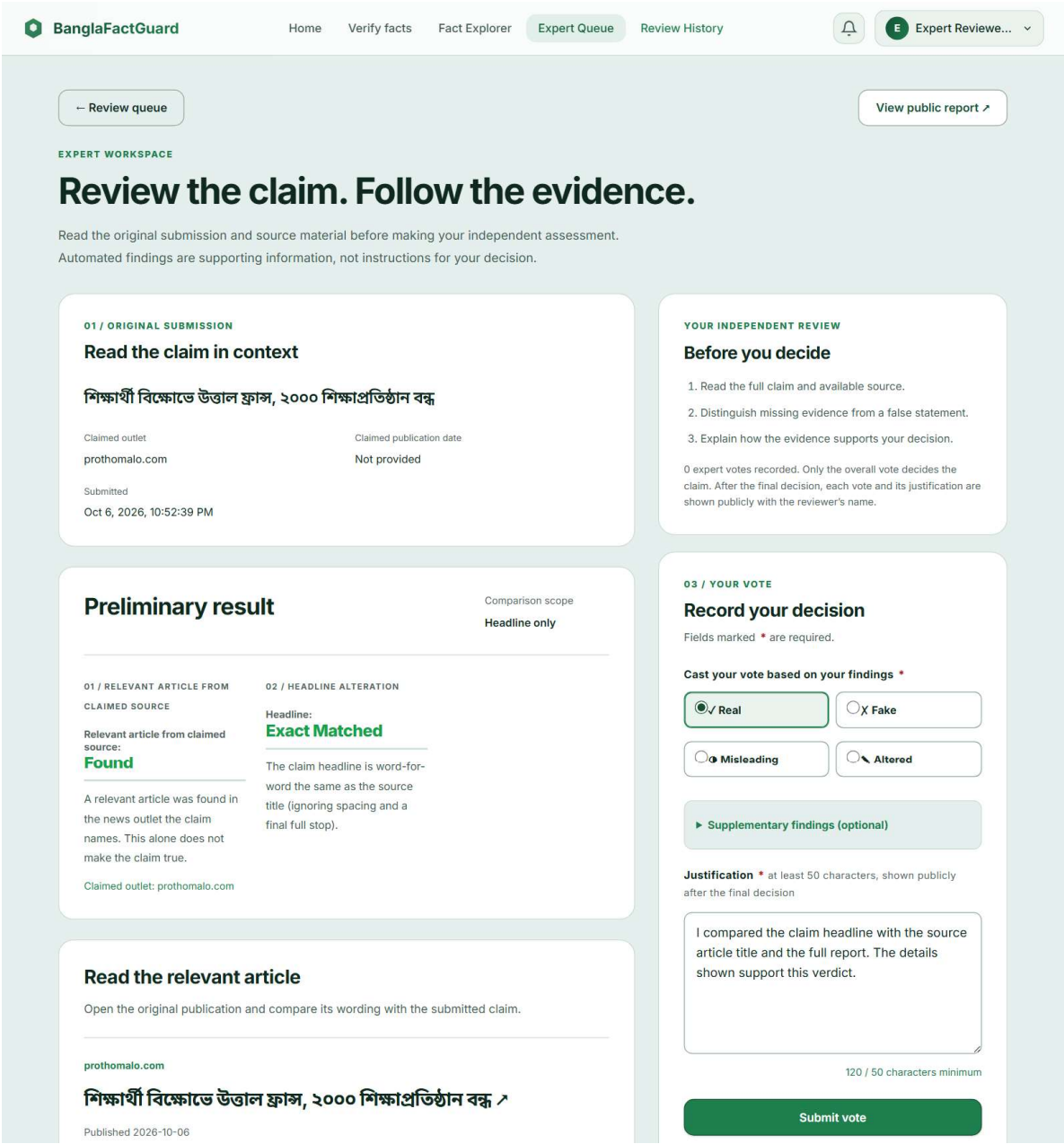


Figure 7.11: VOTE FORM State — Expert Review Page

8. Implementation

8.1 Technology Stack

Table 8.1: Technology Stack

Part	Technology
Website	Angular 19, TypeScript, SCSS, Angular Material/CDK
Backend	Python, FastAPI, SQLAlchemy, Alembic and Pydantic
AI and NLP	PyTorch, timm, Hugging Face Transformers, sentence-transformers, Google Gemini API (photo cards)
Retrieval	HTTP retrieval, browser rendering and article text extraction tools
Data	PostgreSQL 17, Redis 7.2, MinIO
Extension	Preact, Vite 6, Chrome Manifest V3 (side panel and service worker)
Testing	pytest; Karma and Jasmine; Node test runner with simulated browser services

8.2 Code Organisation

The repository separates the backend, website and extension. The backend groups related work into features such as accounts, submissions, verification and expert review. The website follows the main user functions, while shared components provide common forms and result displays. The extension contains the side panel, background processing and page-capture features.

8.3 Key Implementation Details

8.3.1 Authentication and Roles

The system stores password hashes and checks the account role before allowing restricted actions. Expiring access tokens are renewed through a refresh session. The website coordinates simultaneous refresh requests and keeps the session during a temporary network failure. Password recovery uses an emailed code that expires after ten minutes.

8.3.2 Background Jobs and Reuse

Accepted claims and their jobs are saved together. Background workers track progress and recover work abandoned after a restart. Photo card jobs have separate processing capacity because image reading may involve longer waits. Temporary failures can be retried, while permanent failures return an explanation. Compatible saved results can be reused without repeating the full check.

8.3.3 Photo Card Reading

The image-reading service extracts the printed headline, any recognised publisher and any readable date. It does not invent missing values. The extracted claim is checked against the identified outlet or active registered sources when no outlet is identified and broader checking is enabled. An unreadable headline or repeated service failure stops the check with an explanation.

8.3.4 Voting, Escalation and Notifications

The review service prevents repeated votes by the same expert and blocks reviews of the expert's own claims. Automated predictions do not count as expert votes. Unresolved claims are monitored against the configured review limits. Personal notifications are recorded and final-result emails are queued and retried when email delivery is configured.

8.3.5 Browser Extension

The extension stores drafts and activity separately for each account. Its background process submits claims, checks their status and displays available alerts. For screenshots, it crops the captured viewport to the selected area before saving or uploading it. Only an administrator can change the website and service addresses.

8.4 Main API Routes

Table 8.2: Main API Routes (prefix /api/v1)

Operation	Route
Register, login, refresh, logout, profile	POST /auth/register, /auth/login, /auth/refresh, /auth/logout; GET /auth/me
Active verified sources	GET /sources
Queue a text claim; status and result	POST /verify/async; GET /verify/{id}/status, /verify/{id}
Queue a photo card; status and result	POST /photocard/verify/async; GET /photocard/{id}
Queue a text & image claim; result	POST /multimodal/predict/async; GET /multimodal/by-submission/{id}
Submission lookup; own submissions	GET /submissions/{id}; GET /users/me/submissions
Fact Explorer	GET /dashboard/explorer and statistics routes
Expert queue and vote	GET /expert/queue; POST /expert/queue/{id}/vote
Notifications	GET /notifications

9. Testing

9.1 Testing Approach and Strategy

Testing begins with the backend because both the website and browser extension depend on it. The approach starts with individual functions, followed by API routes and the steps used to verify a claim. Each test checks the response and, when needed, the data saved in the database. Simulated Gemini, news search and email services return fixed responses so the tests can be repeated reliably.

The tests cover valid requests, invalid input and service failures, including missing fields, expired sessions, repeated claims, unreadable images and voting ties. Access checks confirm that users can use only the records and actions allowed for their role. The website and extension have separate tests for their behaviour. Screenshots show examples from the application, while model accuracy is evaluated separately from software testing.

Table 9.1 shows the test levels.

Table 9.1: Test levels

Level	What it checks	How it is checked
Unit	Individual functions: headline comparison, date checks, vote weights and token handling	Expected output for a fixed input
Integration	Services working with PostgreSQL, Redis and the job queue	Saved records are checked after the action
API	Every route used by the website and the extension, with valid and invalid requests	Status code, response fields and saved state
Pipeline	The 13 source-verification stages (S01–S13) from claim to saved result	Separate source, headline and date findings
Model	Text & image prediction: output format, repeatability, duplicate detection	Two scores; same input gives the same result
Security	Roles, ownership of records, token and password rules, unsafe input	Request is refused and nothing changes
Non-functional	Concurrent submissions, background recovery, response time	Jobs finish without lost or duplicated records

Each test case uses one of three types of input or checks:

Positive: a valid request that should succeed.

Negative: a request with a missing field, wrong type or unknown ID, access to another user's record or a service failure. The request should be rejected with a clear message.

State: a check that the action changes the saved data as expected.

9.2 Test Tools

Table 9.2: Test tools

Item	Tool
Backend unit and API tests	pytest
Database	Local PostgreSQL test database
API calls	HTTP test client (needed for the integration suite)
Simulated external services	Simulated Gemini, search and email responses
Model tests	Saved model files; two real-model tests are disabled by default

Command used to run the backend tests: python -m pytest tests/unit -q --no-cov

9.3 Item Pass/Fail Criteria

A test case passes only when it meets every criterion for its item class. If a required resource is not enabled, the case is recorded as skipped and counts as neither a pass nor a failure.

Table 9.3: Pass and fail criteria by item class

Item class	Passes when	Fails when
Function	Returns the expected value for every fixed input	An incorrect value or an unexpected error
API route	Correct status code, correct response fields and correct saved state	Wrong status, missing field or a change that did not happen
Access control	An unauthorised request or a request for another user's record is rejected; the record stays unchanged	The request succeeds or changes the record
Verification result	Source, headline and date findings are separate; missing evidence is shown as Incomplete or Not Found	A missing source is labelled Fake or an unavailable score is shown as 0
Voting	Claim is final only when all voting rules are met	The claim is finalized too early or while the vote is tied
Background job	Job finishes or resumes after a restart, with one saved result	A job is lost, a result is duplicated or a failure has no reason
Model output	Two scores for Real and Fake; same input gives the same result	Wrong format or different results for the same input

Passing these cases shows that the tested software rules work as expected. It does not establish model accuracy or production readiness.

9.4 Summary of Executed Test Runs

The backend unit test suite ran 426 tests: 424 passed, 0 failed and 2 were skipped.

Table 9.4: Executed automated test run

Suite	Tests	Passed	Failed	Skipped	Time
Backend unit tests (44 files)	426	424	0	2	20.4 s

The two skipped tests require local model weights and the real-model setting, which was not enabled. They are not counted as passed.

9.5 Test Cases and Outcomes

The following tables group test cases by feature. Pass means the case was executed in the automated run shown in Table 9.4.

9.5.1 Authentication, Accounts and Access Control

Table 9.5: Test cases — authentication, accounts and access control

ID	Scenario	Input / steps	Expected result	Status
TC-01	Refresh access token	Expired access token, valid refresh token	A new token is issued; the user stays signed in	Pass
TC-02	Lost refresh response	Refresh succeeds but the reply is lost	User is not logged out within the grace window	Pass
TC-03	Logout	Sign out, then use the old refresh token	Token is revoked immediately	Pass
TC-04	Register with valid details	Valid name, email and password	Account created; password stored as a hash; no password in the response	Pass
TC-05	Email already in use	Register again with the same email	Rejected (409); no second account	Pass
TC-06	Short password	Password shorter than 8 characters	Rejected (422); no account created	Pass
TC-07	Wrong password and unknown email	Login with each	The same error message is returned for both, so registered emails cannot be identified	Pass
TC-08	Password reset code	Use a valid code, then use the same code again	Password changed once; second use rejected	Pass
TC-09	Expired reset code	Use a code after its time limit	Rejected; password unchanged	Pass
TC-10	No token	Call GET /users/me/submissions without a token	Rejected (401)	Pass
TC-11	Expired or changed token	Send an expired token, then a token with edited content	Both rejected (401)	Pass
TC-12	User calls an expert route	Normal user calls GET /expert/queue	Refused (403)	Pass
TC-13	Expert calls an admin route	Expert tries to change voting rules or sources	Refused (403); nothing changed	Pass
TC-14	Own records only	User A requests submissions and notifications	Only user A's records are returned	Pass

9.5.2 Text & Source Verification

Table 9.6: Test cases — text & source verification

ID	Scenario	Input / steps	Expected result	Status
TC-15	Matching claim, full workflow	Test claim with a matching source article	Source and headline match; evidence and findings are saved	Pass
TC-16	Wrong claimed date	Correct headline, wrong date	Headline Matched, date Mismatched	Pass
TC-17	Changed number	Headline with an altered number	Headline Altered	Pass
TC-18	All search providers fail	Search clients raise errors	Search is reported as Incomplete	Pass
TC-19	Unknown outlet, broader checking off	Unrecognised outlet with the fallback option disabled	Rejected before a submission is created	Pass
TC-20	Repeated claim	Same claim submitted twice	A compatible saved result is reused	Pass
TC-21	Incomplete result not reused	Earlier result was Incomplete	Claim is queued again, not served from the saved result	Pass
TC-22	Body similarity scores	Claim with article text	Four scores; an unavailable score is never shown as 0	Pass
TC-23	No source found	Search finishes with no relevant article	Result is Not Found; no automated Fake verdict	Pass
TC-24	Negation	"X approved" checked against "X not approved"	Headline Altered	Pass
TC-25	Changed person or organisation	Same headline with a different name	Headline Altered	Pass
TC-26	Bangla and English digits	Same number written in Bangla and in English digits	Treated as the same number; Headline Matched	Pass
TC-27	Off-source result	Search returns a page from a website that is not allowed	Page rejected; not used as evidence	Pass
TC-28	Missing claimed date	Claim submitted without a date	Date finding shown as unavailable, not Mismatched	Pass
TC-29	Empty headline	Empty or spaces-only headline	Rejected (422); no submission created	Pass
TC-30	Unknown submission ID	GET /verify/{id}/status with a random ID	Not found (404)	Pass

9.5.3 Photo Card and Text & Image Verification

Table 9.7: Test cases — photo card and text & image verification

ID	Scenario	Input / steps	Expected result	Status
TC-31	Accepted before processing finishes	Upload a photo card	Request accepted before image reading finishes; processing continues in the background	Pass

ID	Scenario	Input / steps	Expected result	Status
TC-32	Image-reading service unavailable	All image-reading requests fail	Failure message shown; no verdict	Pass
TC-33	Unreadable claim or unknown outlet	No readable headline or no recognised outlet while broader checking is disabled	Clear message explains why processing stops	Pass
TC-34	Crash recovery	Worker stops after the extracted claim is saved	A new worker resumes from the saved extraction	Pass
TC-35	Invalid file type	PDF uploaded as an image	Rejected (415) before acceptance	Pass
TC-36	Empty article text	Spaces-only article text with an image	Rejected (422)	Pass
TC-37	Duplicate text and image	Same text and image again	Earlier prediction is reused after the similarity check	Pass
TC-38	Different image	Same text, different image	Not treated as a duplicate	Pass
TC-39	Image too large	Upload an image above the size limit	Rejected before processing	Pass
TC-40	Corrupted image	A damaged file named .jpg	Rejected with a clear message; no server error	Pass
TC-41	Photo card without date or outlet	Card shows only a headline	Missing date and outlet fields stay empty; no values are guessed	Pass
TC-42	Text & image without an image	Headline and text, no image	Rejected (422)	Pass
TC-43	Prediction format	Valid text and image	Two scores (Real, Fake) that add up to 1; label Likely Real or Likely Fake; shown as preliminary	Pass
TC-44	Same input, same result	Run the same input twice with result reuse off	Same label and scores both times	Pass
TC-45	Long article text	Article longer than 128 tokens	Processed without error; text beyond the limit is truncated	Pass

Section 5.1.2 reports the model evaluation on 960 test items, with 80% accuracy. This is a model result, so it is not repeated as a software test result here.

9.5.4 Expert Review, Voting, Administration and Notifications

Table 9.8: Test cases — expert review, voting, administration and notifications

ID	Scenario	Input / steps	Expected result	Status
TC-46	All rules met	Required voters, leading weight and lead margin are all satisfied	Claim is finalized	Pass

ID	Scenario	Input / steps	Expected result	Status
TC-47	One rule not met	The threshold, required number of voters or lead margin is not met	Claim stays open	Pass
TC-48	Tie	Equal top weights	Never finalized; escalated at the vote limit	Pass
TC-49	New expert weight	Expert has fewer completed reviews than required	Neutral weight of 1	Pass
TC-50	Own claim and double vote	Expert votes on own claim; expert votes twice	Both rejected	Pass
TC-51	Review time limit	The configured time limit passes without a decision	Claim escalated; admin notified once	Pass
TC-52	Admin rights	Admin opens escalated and normal claims	Can decide escalated claims; view-only for others; decision is final	Pass
TC-53	Personal notifications	Check each method for users, experts and admins	Each preliminary and final update is recorded once	Pass
TC-54	Guest submitter	Claim submitted without an account	No account notification or email	Pass
TC-55	Email server failure	Mail server returns an error	Email is retried; no extra in-app notification	Pass
TC-56	Short justification	Vote sent with fewer than 50 characters	Rejected (422); vote not saved	Pass
TC-57	Automated prediction is not a vote	Claim has a Likely Fake prediction and no expert votes	Prediction is not counted in the vote total	Pass
TC-58	Vote weight is kept	Expert votes, then the expert's credibility tier changes	The saved vote keeps the weight used at voting time	Pass
TC-59	Invalid voting rules	Admin saves 0 required voters or a negative margin	Rejected; old rules stay active	Pass
TC-60	Inactive source	Admin deactivates a source, then a claim names that outlet	Source is not searched	Pass

9.5.5 Data Integrity, Reliability, Security and Performance

These cases cover the non-functional requirements NFR-1, NFR-3, NFR-6 and NFR-7. No results have been recorded for them yet.

Table 9.9: Test cases — data integrity, reliability, security and performance

ID	Scenario	Input / steps	Expected result	Status
TC-61	Submission and job saved together	Force the job record to fail while saving a claim	No submission is left without a job	Pass
TC-62	Same claim at the same time	Send two identical claims together	One result record; both submissions point to it	Pass

ID	Scenario	Input / steps	Expected result	Status
TC-63	Text job after restart	Stop the worker in the middle of a text check, then restart	Job resumes and finishes with one saved result	Pass
TC-64	Temporary and permanent failures	Temporary search error, then an unreadable input	Temporary error is retried; permanent failure shows a reason	Pass
TC-65	SQL text in a headline	Headline containing '; DROP TABLE users; --	Saved as plain text; database tables unchanged	Pass
TC-66	Script text in a headline	Headline containing a <script> tag	Shown as plain text; not run	Pass
TC-67	Fact Explorer data	Open the Explorer routes without signing in	Only available reports are listed; email addresses and personal data are excluded	Pass
TC-68	Ten checks at once	Send 10 text claims together	All are accepted; all finish; none fail or are lost	Pass
TC-69	Submit response time	Send text, photo card and text & image claims	Each request is accepted within a few seconds, before processing ends	Pass

9.6 Observed Performance

No formal load or stress test has been recorded. A timing log of eight source checks shows processing times of about 82–153 seconds. These times cover pipeline processing only and exclude the wait for expert review. The sample is small and does not guarantee response times. TC-68 and TC-69 are planned to measure concurrent processing and submission response times.

9.7 Tests Not Executed

Table 9.10: Tests not executed

Item	Reason	Status
Separate backend integration suite	Requires a working test database and a compatible HTTP test client	Not executed
Real-model tests (2)	Require local model weights and explicit activation	Skipped
Accuracy of source verification	No labelled Bangla test set	Not measured
Load and stress testing	No load-test result has been recorded	Not executed
Cases marked To run in Section 9.5	Planned, but results have not been recorded	Not executed

9.8 Risks and Contingencies

Table 9.11: Testing risks and contingencies

Risk	Contingency
Gemini quota or network errors stop photo card checks	Retry temporary failures and explain when image reading remains unavailable
A news website blocks requests or changes its layout	Try other retrieval methods and report incomplete evidence if access fails
Real-model behaviour differs from simulated test responses	Evaluate real models with labelled data and retain expert review
Processing is interrupted by a restart	Save accepted jobs and resume from the saved state
Too few experts to reach agreement	Use configurable review limits and escalate to an administrator
A test changes real data	Use a separate test database and generated test accounts
A test itself is wrong	Reproduce the failure manually before treating it as a software defect

10. User Manual

This chapter explains how to submit a claim, read a result and manage account activity. It also covers expert and administrator tasks. The interface uses English, while the verification forms accept Bangla content.

10.1 Getting Started

Open the website. The home page introduces the three verification methods (Figure 10.1). Guests can submit claims; signing in links submissions to an account and enables personal notifications.

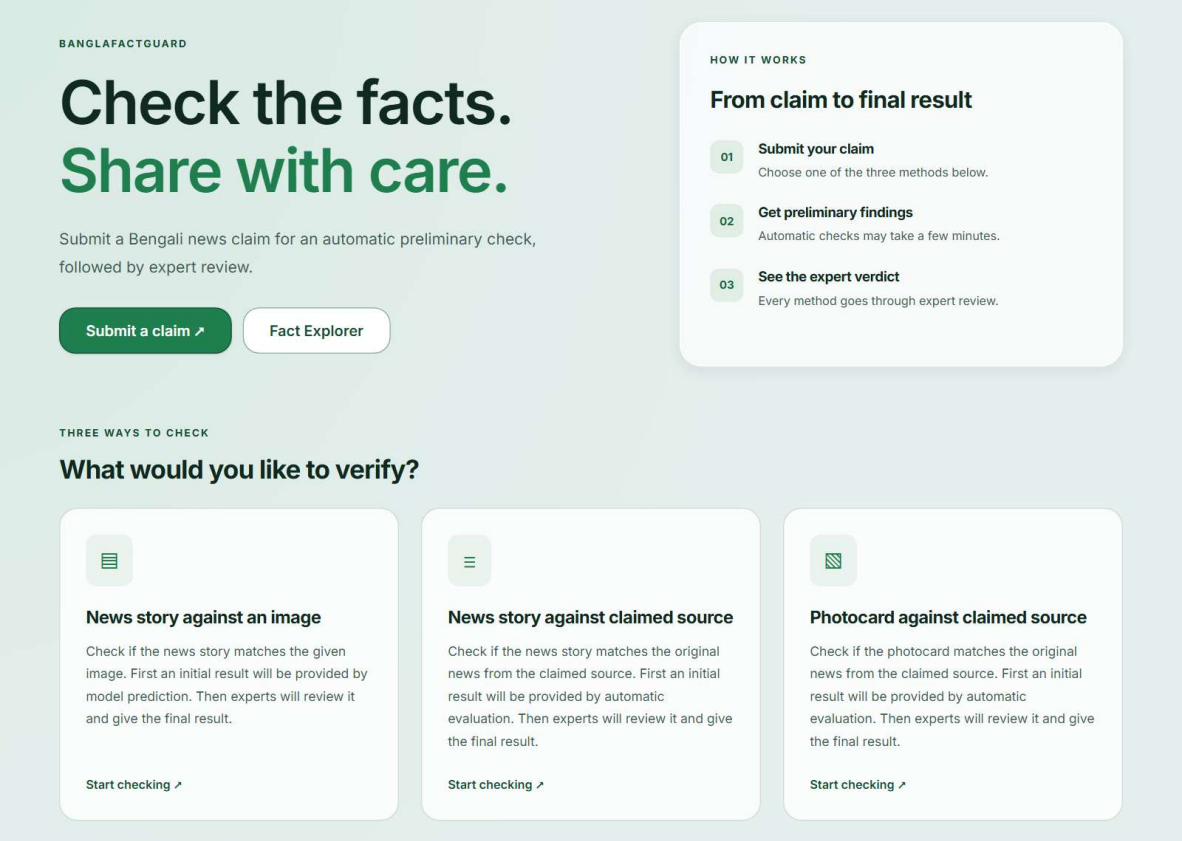


Figure 10.1: Home Page

10.2 Creating an Account and Signing In

1. Click **Get Started** (or **Login**, then **Create one**).
2. Enter your full name, email and a password of at least 8 characters with an uppercase letter and a number. Confirm the password and click **Create Account** (Figure 10.2).
3. To sign in later, click **Login**, enter your email and password and click **Sign In**.
4. If you forget your password, click **Forgot password?**, enter your email, then type the code sent to your email and a new password.

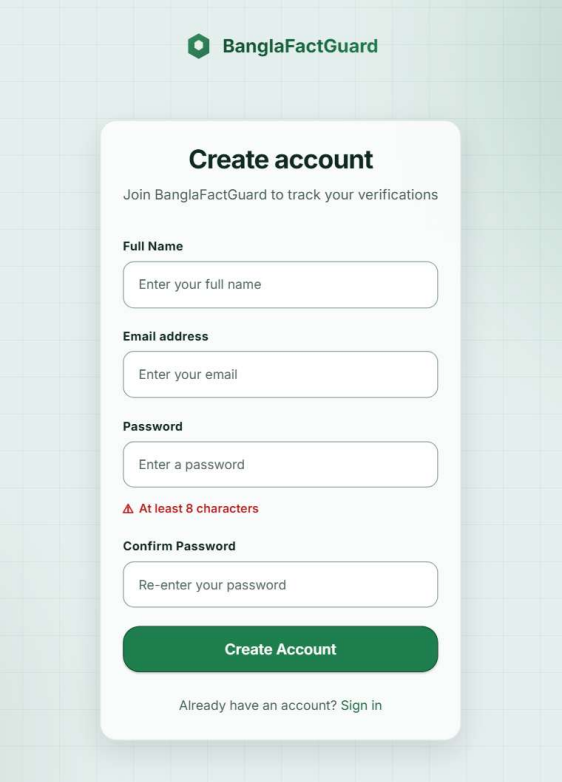


Figure 10.2: Create Account Page

10.3 Checking a Claim

Click **Verify facts** and choose a method card. You can hide the card descriptions with **Hide descriptions**.

10.3.1 Text & Source (News story against claimed source)

1. Type or paste the Bangla headline.
2. Choose the publisher named in the claim when it is known. If no active publisher is identified, the current system can search across registered sources.
3. Optionally paste the article text and choose the claimed publication date.
4. Click **Verify news**. The page shows "Verification in progress" (Figure 7.6). You may leave the page; the check continues.

Figure 7.5 shows the form. Completion time varies with evidence retrieval, model processing and server load.

10.3.2 Photo Card (Photocard against claimed source)

1. Click the upload box and choose the card image (JPEG, PNG, WebP or GIF, up to 10 MB).
2. Check the preview, then click **Verify photo card**.
3. The system reads the headline and any visible outlet and date. It then searches the identified outlet or the active registered sources. If the headline cannot be read or the service fails, the result explains the problem and offers another image upload (Figure 7.9).

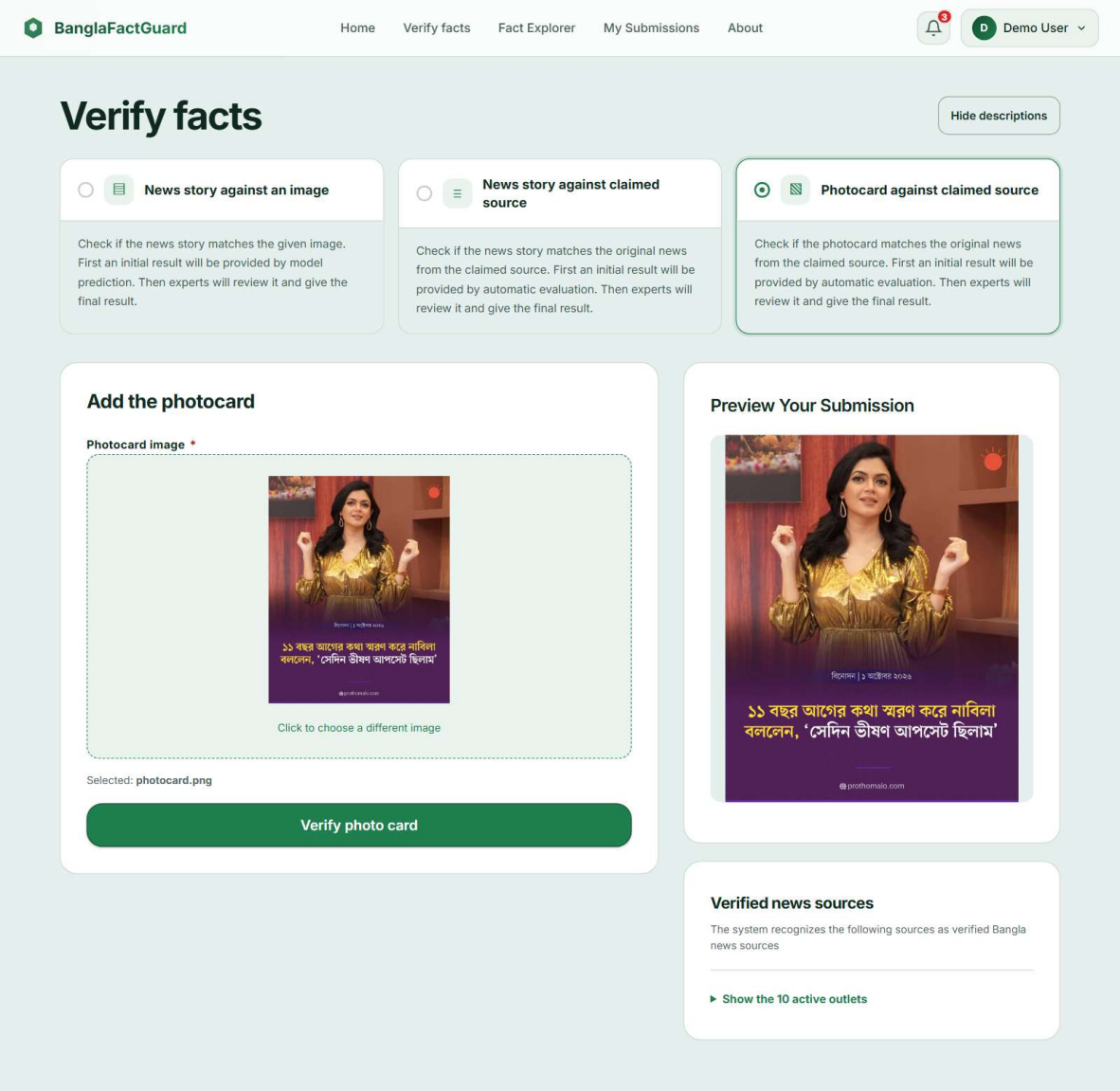


Figure 10.3: Photo Card Form with Preview

10.3.3 Text & Image (News story against an image)

1. Enter the headline and the article text.
2. Upload the news image and check the preview.
3. Click **Check text and image**. The result shows **AI decision: Likely Real** or **Likely Fake**, which is a preliminary estimate only.

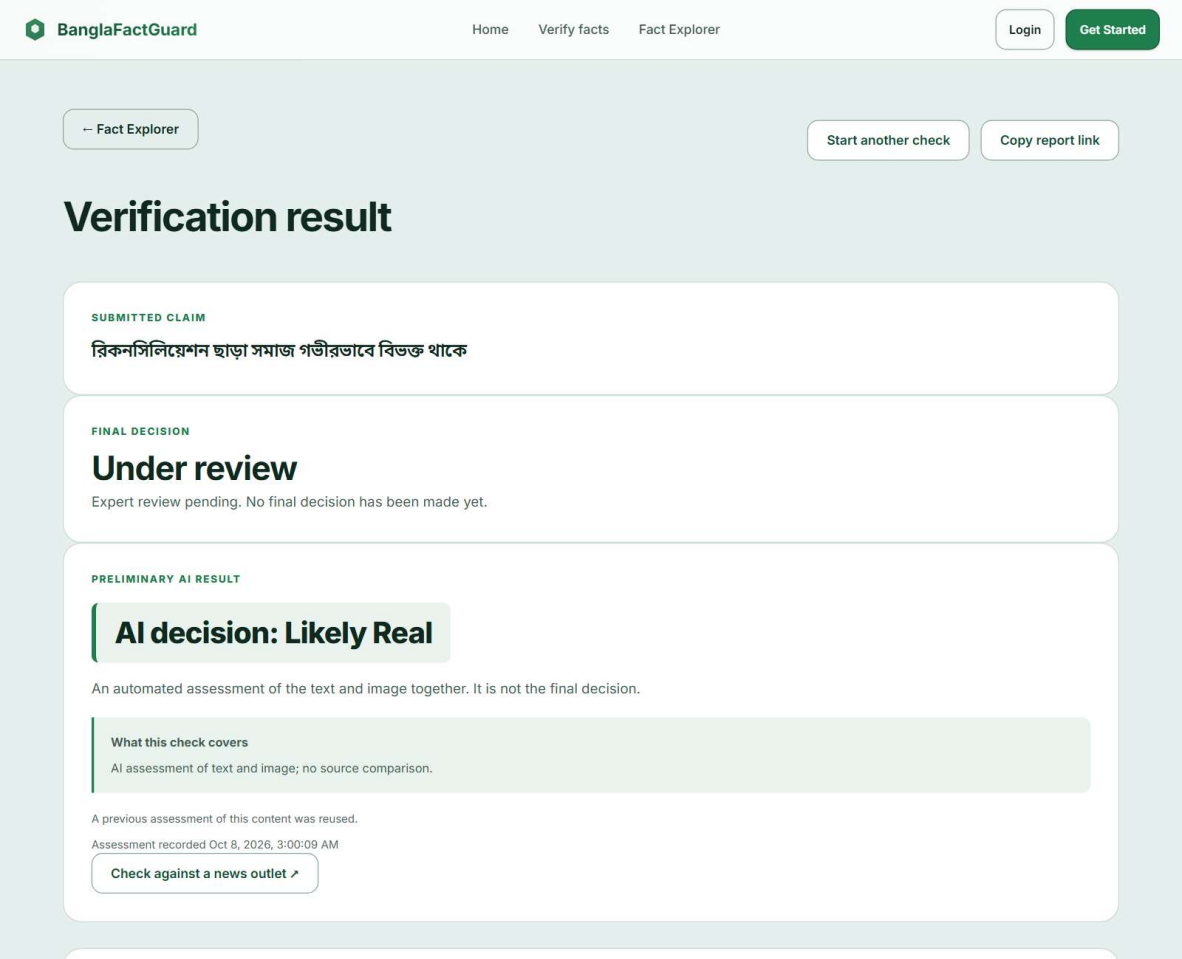


Figure 10.4: Text & Image Result with a Reused Assessment

10.4 Reading a Result

The report shows the claim, the review status and the available preliminary findings. Source-based checks also show relevant articles and comparisons when available. Under review means that no final verdict has been published. A completed review adds the final decision and voting details.

- Source findings show whether a relevant article was found within the stated search scope. Not Found does not mean the claim is false and Incomplete means the check could not reach a conclusion.
- Headline findings show an exact match, preserved meaning or an alteration when a comparison is possible. Meaningful differences are listed beside the claim and source wording.
- Date findings compare a supplied or extracted date with the article date. If either is unavailable, a definite match or mismatch cannot be shown.
- Body similarity scores appear when article text is supplied. They measure text overlap and meaning similarity, not the probability that a claim is true.

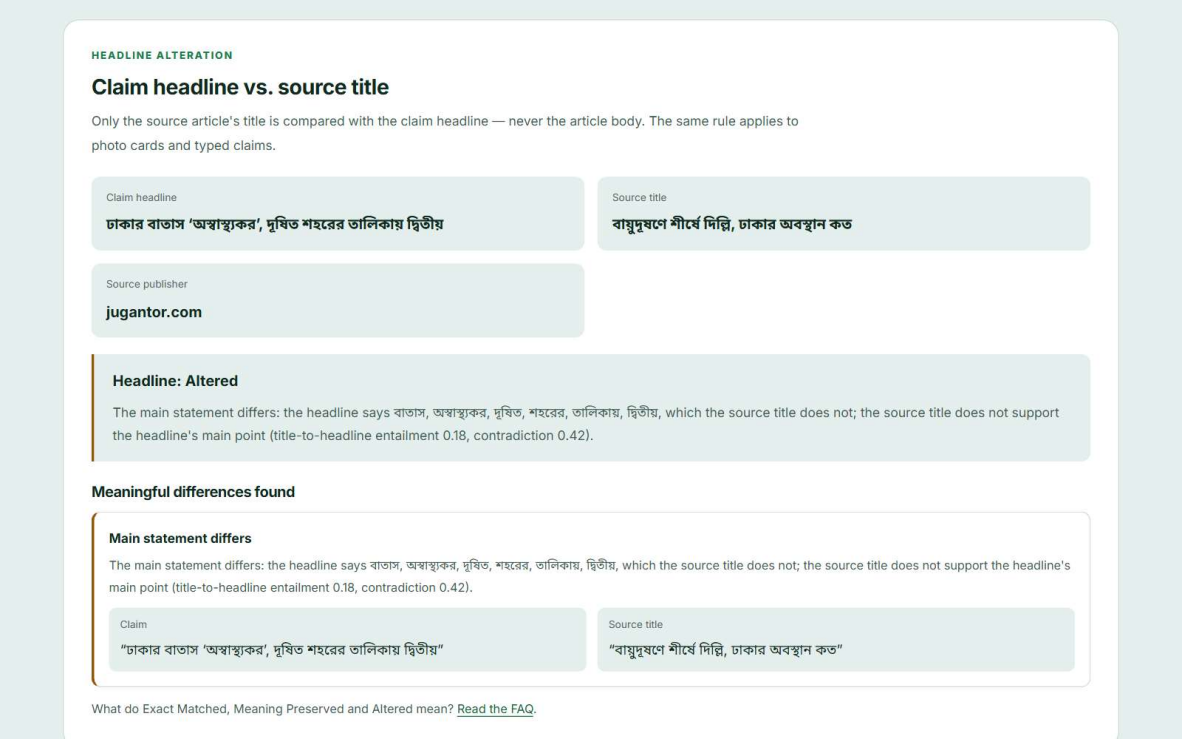


Figure 10.5: Headline Comparison Showing an Altered Headline

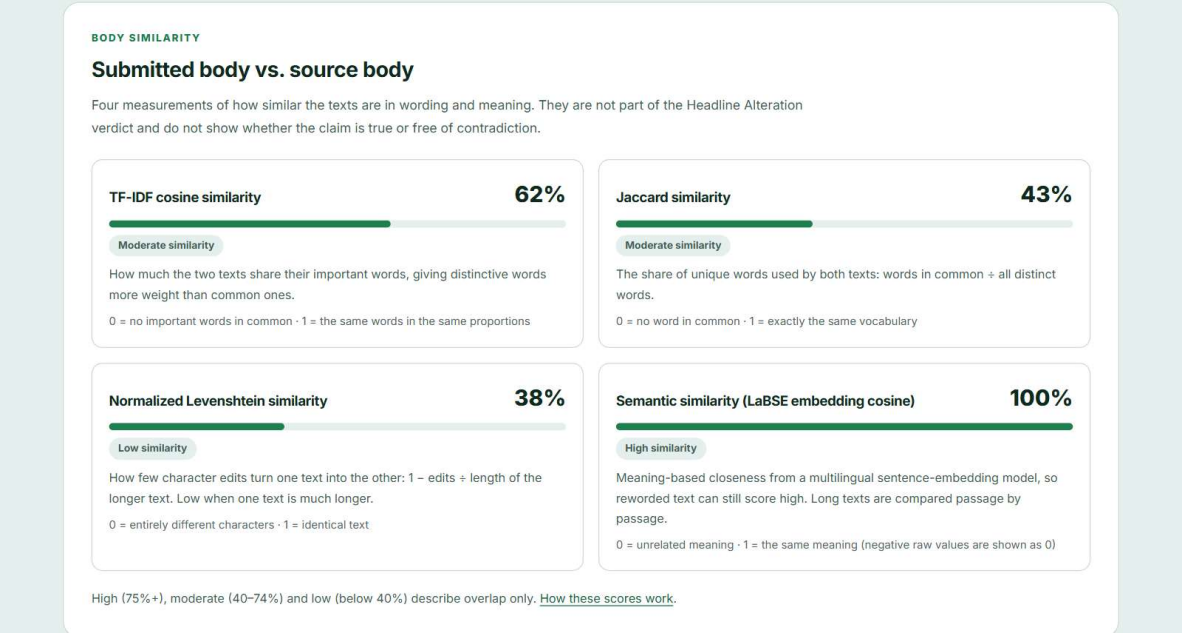


Figure 10.6: Body Similarity Scores

Use **Copy report link** to share a result and **Start another check** to begin again.

10.5 Fact Explorer

Click **Fact Explorer** to search previous claims (Figure 10.7). The top shows archive counts. Choose **Final decisions** or **Under review**, type a keyword, choose a decision and content type and click **Search**. **More filters** adds date and outlet filters. Click **Read full report** to open a claim.

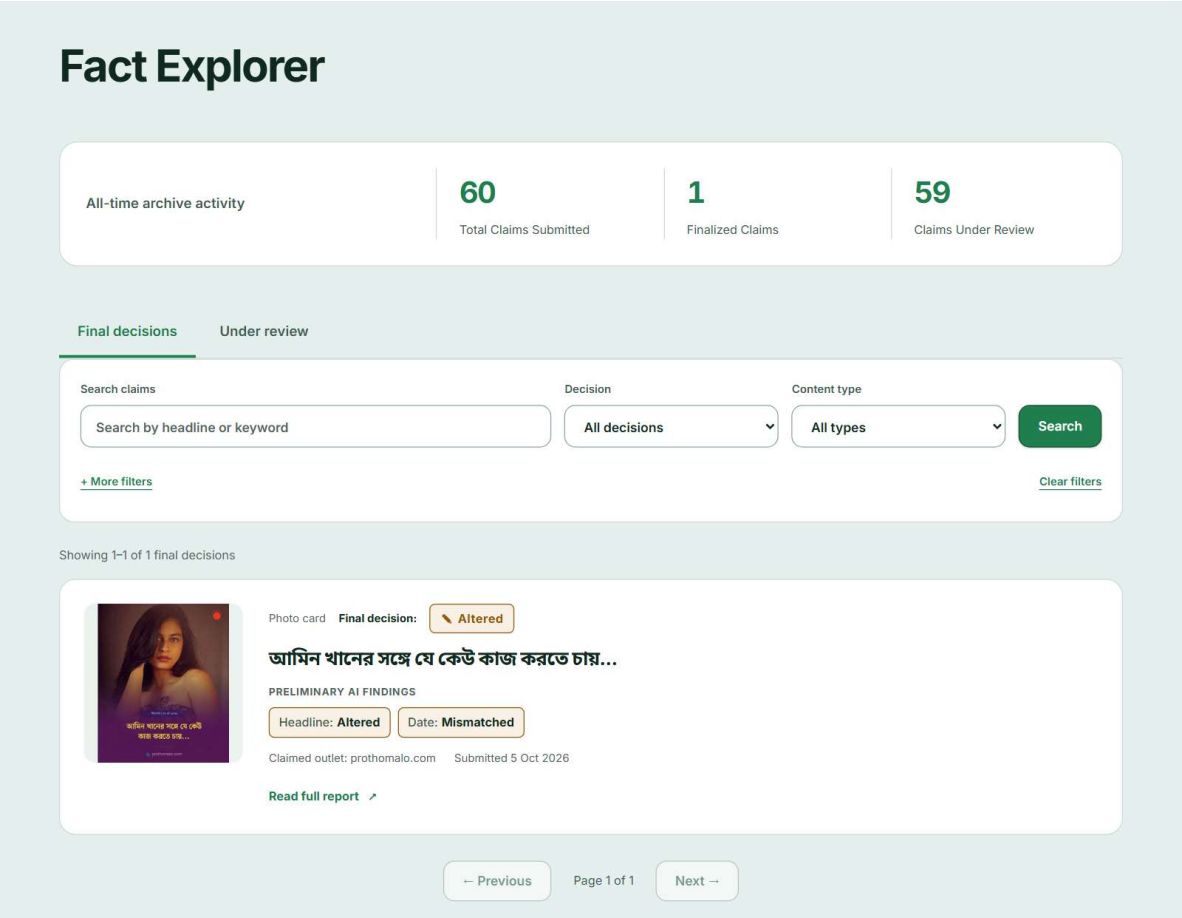


Figure 10.7: Fact Explorer

10.6 My Submissions, Notifications and Settings

My Submissions lists your claims (Figure 10.8). The bell opens result notifications (Figure 10.9); marking them as read keeps them in the list. Final results are emailed when delivery is configured. Account settings lets you change your name and password.

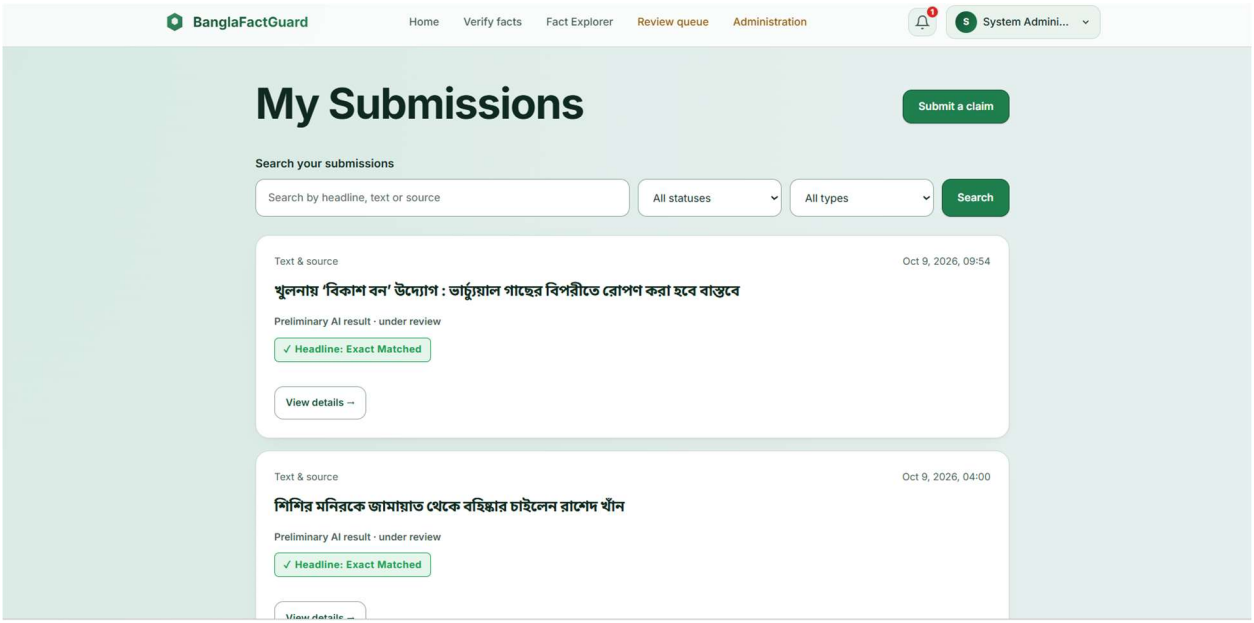


Figure 10.8: My Submissions

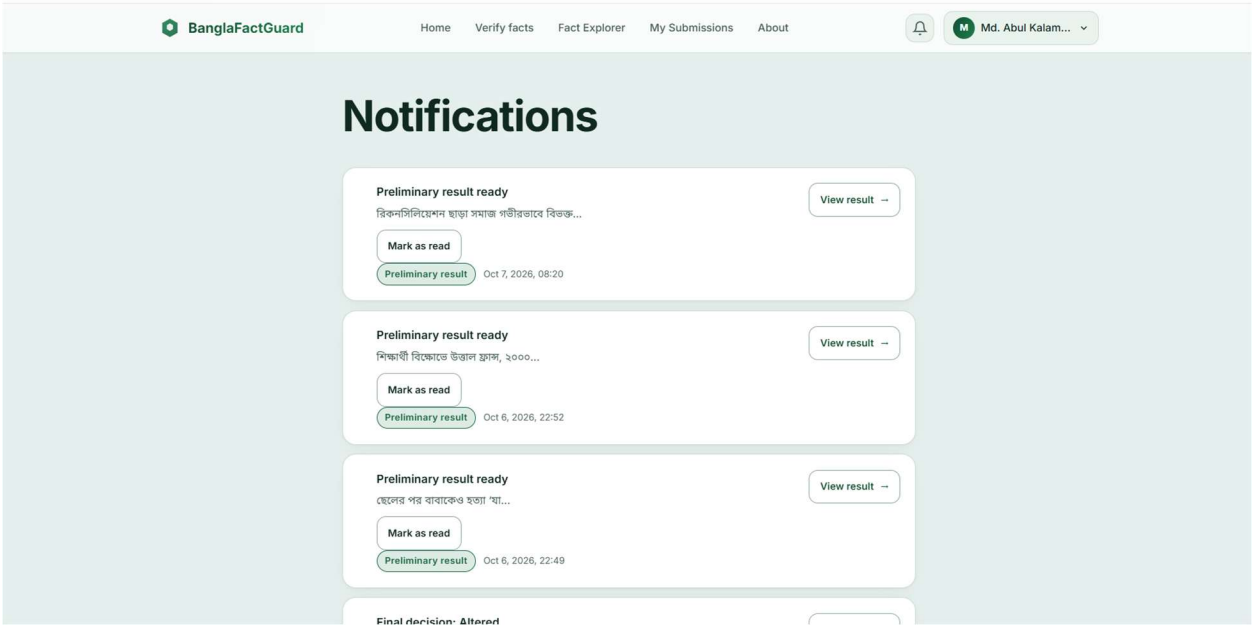


Figure 10.9: Notifications

10.7 Using the Browser Extension

1. Pin **BanglaFactGuard — Browser Extension** in Chrome and click its toolbar icon on the news page. The side panel opens.

2. Choose **Text & source**, **Photo card** or **Text & image**.
3. Select text on the page and click **Use selected text** or type it. For images, click **Upload image** or **Select screenshot area** and drag over the part of the page you want.
4. Adjust the image with **Apply crop**, **Reset crop** or **Remove**.
5. Click Submit for verification. Activity first shows the claim being checked and then the available result. Desktop alerts depend on browser and notification settings. Use View details to open the full website report.
6. Sign in separately through the Account tab using your website account. Switch the extension off to pause new activity and alerts; accepted checks continue on the server.

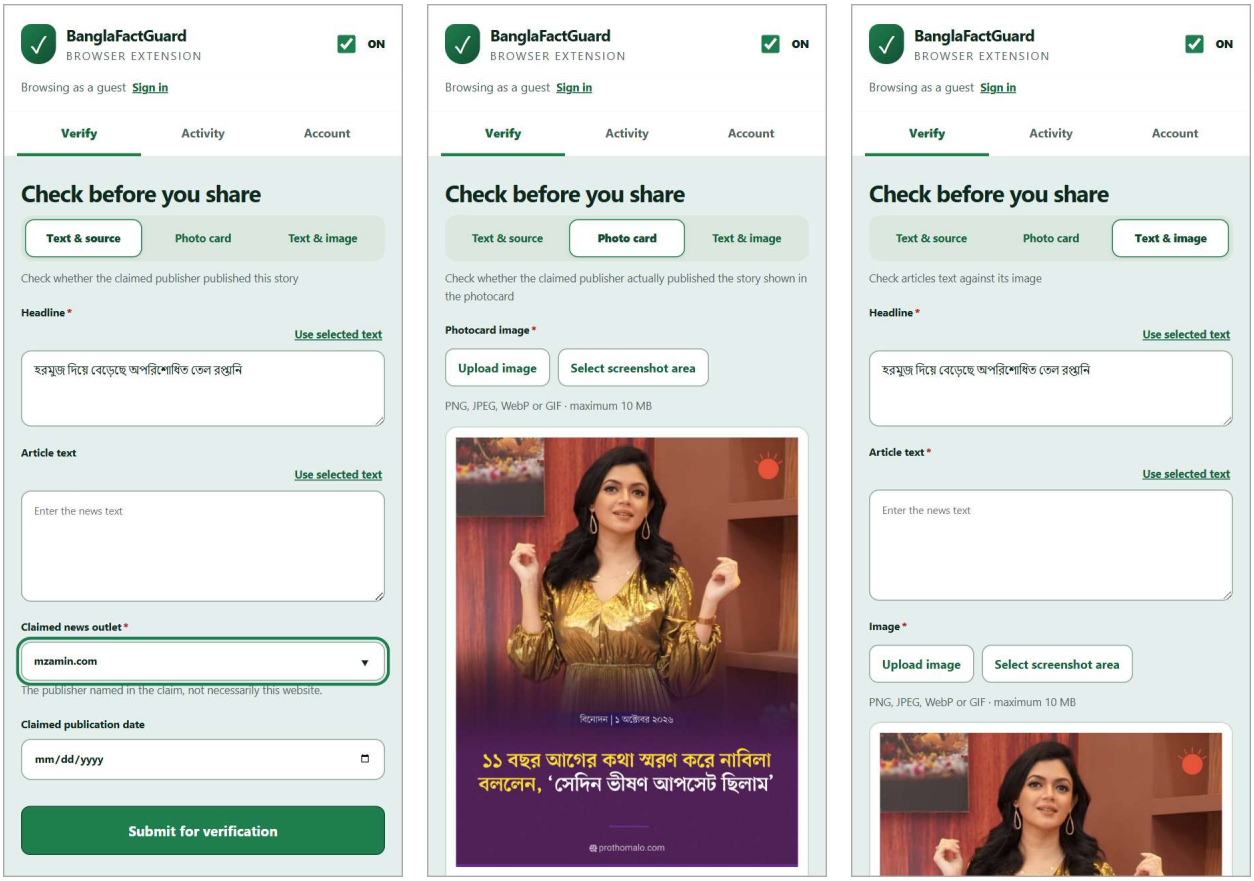


Figure 10.10: Extension Verify Tab — Text & Source, Photo Card and Text & Image

The account and activity screens are shown in Figure 7.10.

10.8 Expert Tasks

1. Sign in with the account created by the administrator. Open **Expert Queue** (Figure 10.11).
2. Search if needed and click **Review & vote** on a claim.
3. Read the claim, the preliminary findings and the article (Figure 7.11). Remember that automated findings are supporting information, not instructions.
4. Choose **Real**, **Fake**, **Misleading** or **Altered**, optionally add supplementary findings, write a justification of at least 50 characters and click **Submit vote**.
5. Open Review History to see your votes and available final decisions. Account settings shows your review record. New experts use the default vote weight until they reach the required number of completed reviews.

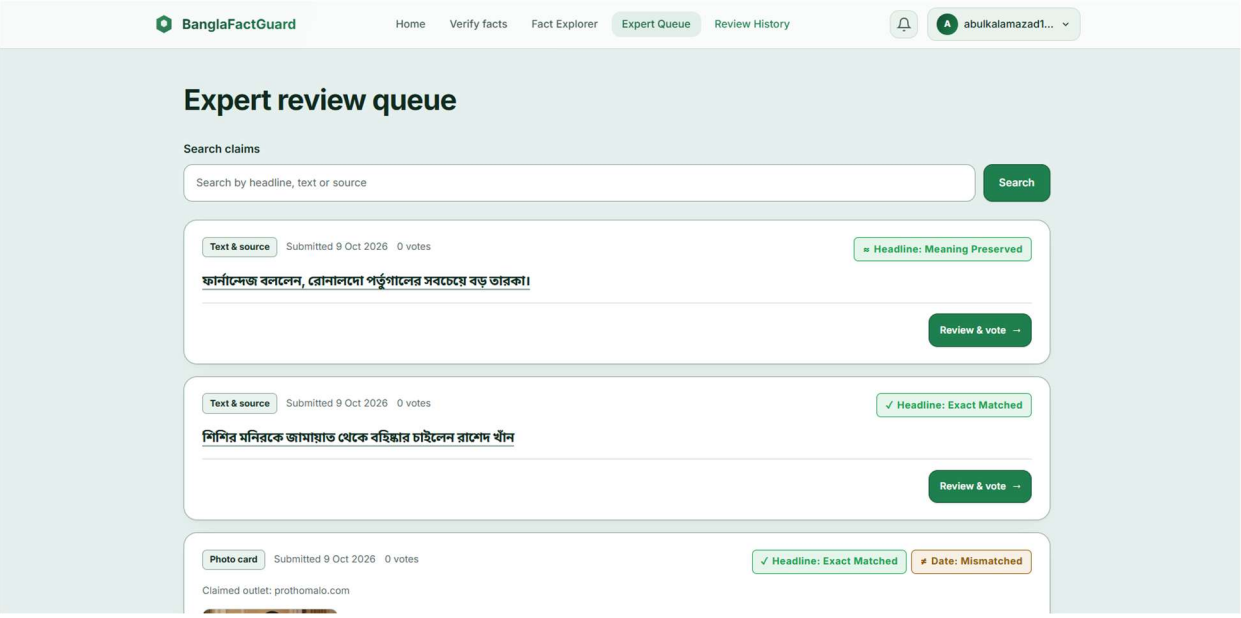


Figure 10.11: Expert Review Queue

10.9 Administrator Tasks

The **Admin dashboard** (Figure 10.12) shows escalated claims that need your decision, activity counts, claims waiting longest and recent decisions. Use the **Experts**, **Sources** and **Voting & tiers** buttons to manage the system.

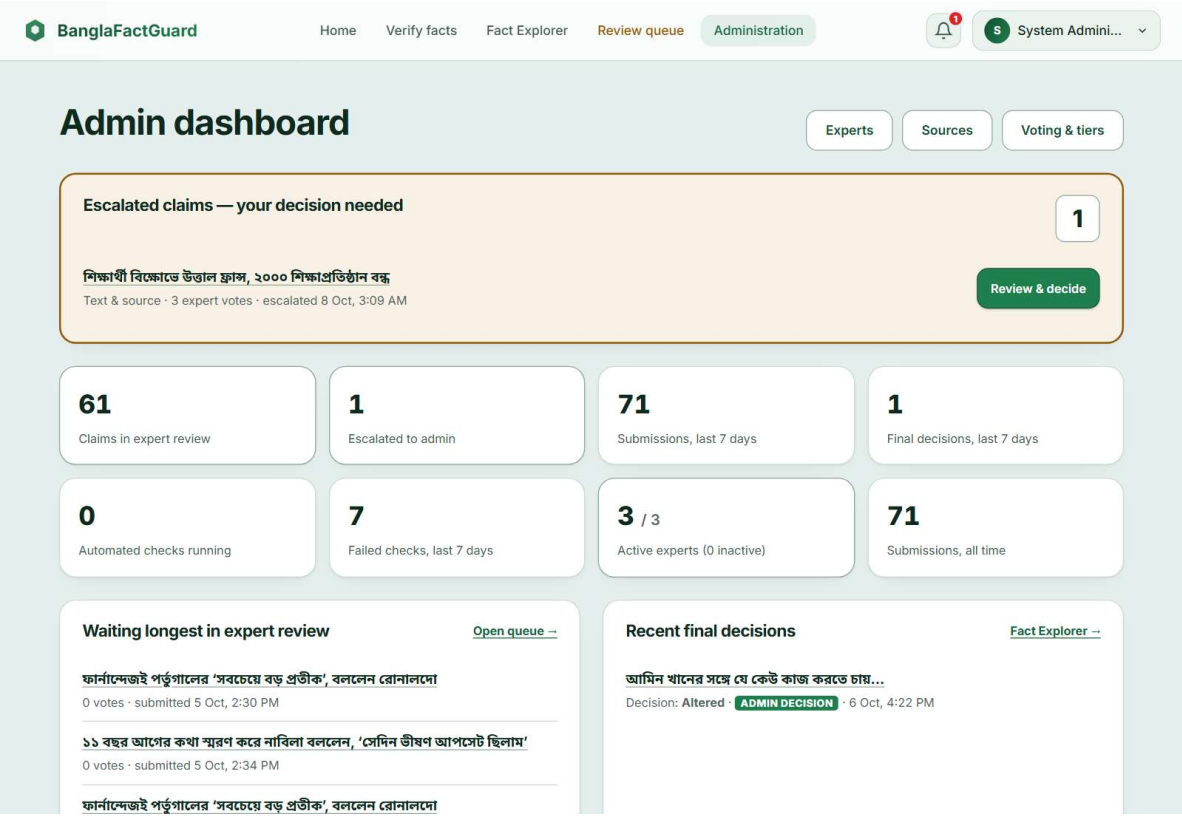


Figure 10.12: Administrator Dashboard

- **Experts:** click **Create expert**, enter name, email, password and expertise area and click **Create Expert Account**. Use **Edit**, **Deactivate** and **Reset password** in the table (Figure 10.13).
- **Sources:** click **Register New Source** to add an outlet or use **Edit**, **Disable** and **Delete** on a card (Figure 10.14). Only active sources are used for verification.
- **Voting & tiers:** set Min voters (M), Verified threshold (T), Lead margin, Activation threshold (N), Max review votes and Max review hours, then **Save Configuration**. Add or edit tiers with their accuracy range and weight (Figure 10.15).
- **Escalated claims:** in **Review queue**, choose **Escalated — needs your decision**, click **Review & decide**, choose the verdict, explain it and submit. Your decision is final; the public result then shows it with Voting Details (Figure 10.16). Claims that are not escalated are view-only.

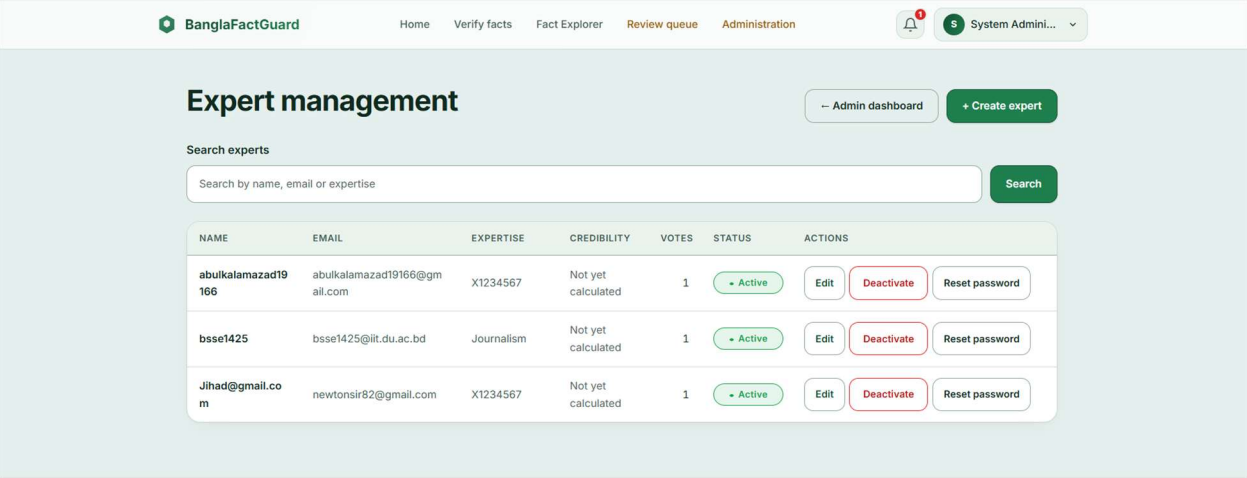


Figure 10.13: Expert Management

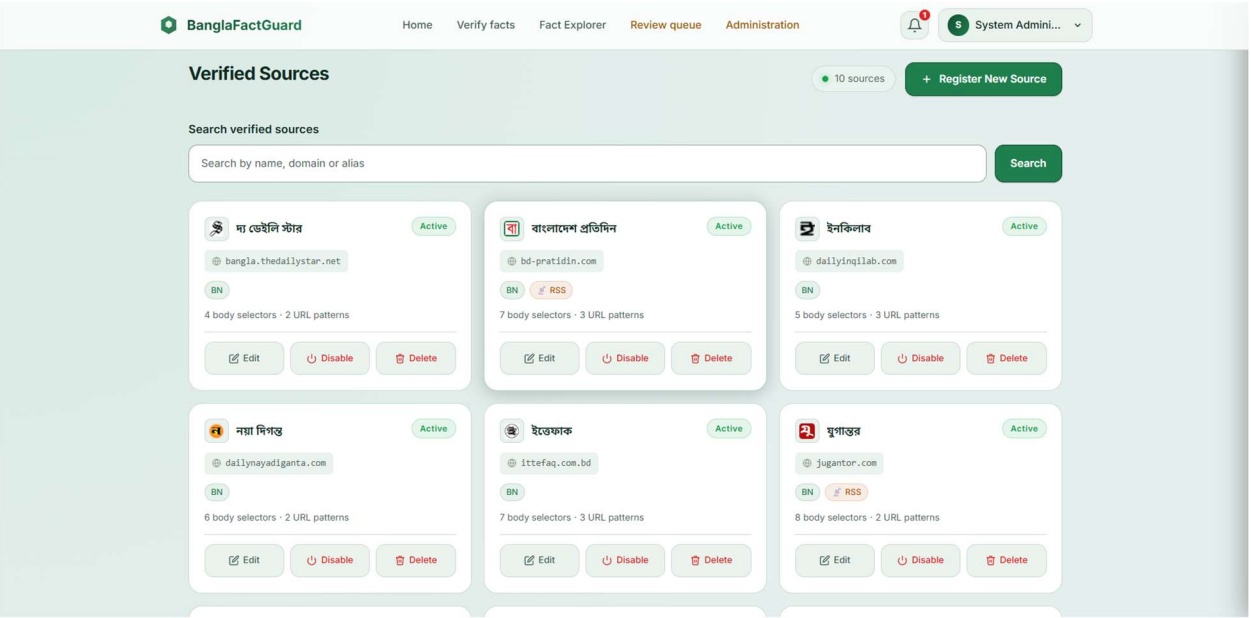


Figure 10.14: Verified Sources

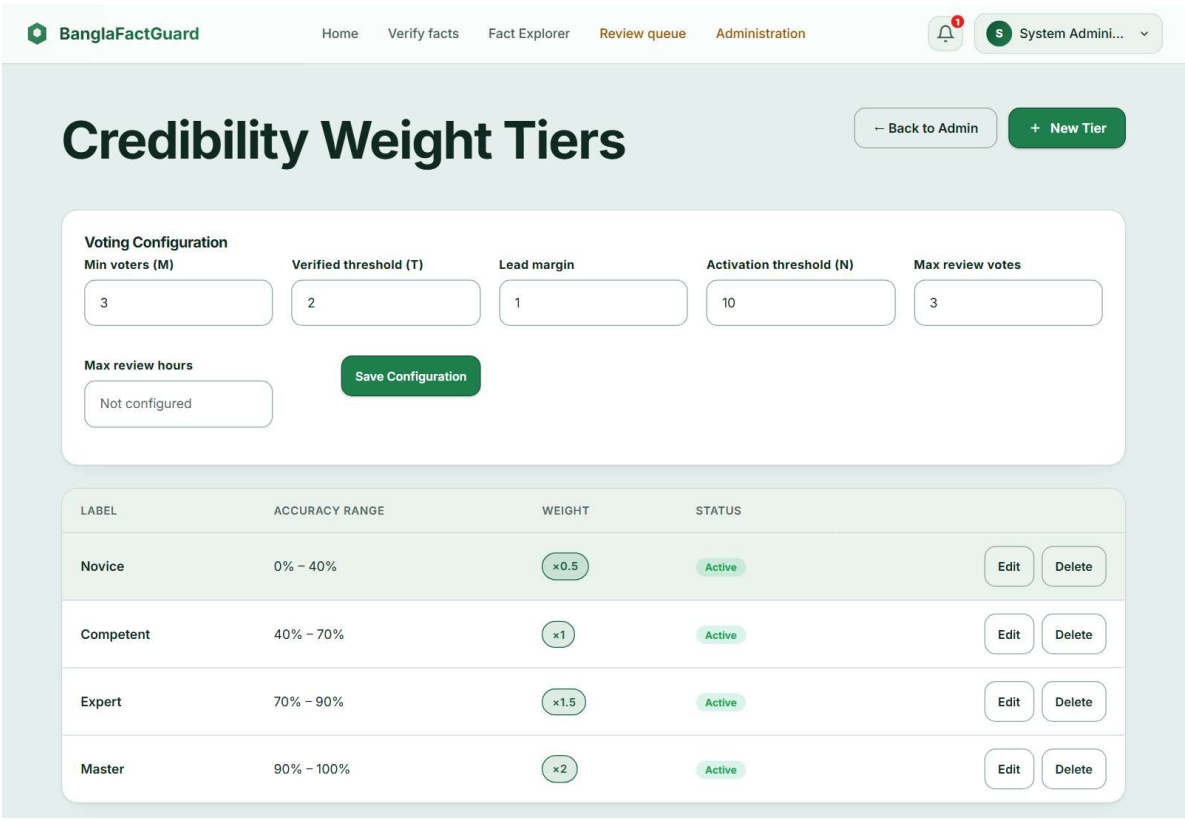


Figure 10.15: Voting Configuration and Credibility Weight Tiers

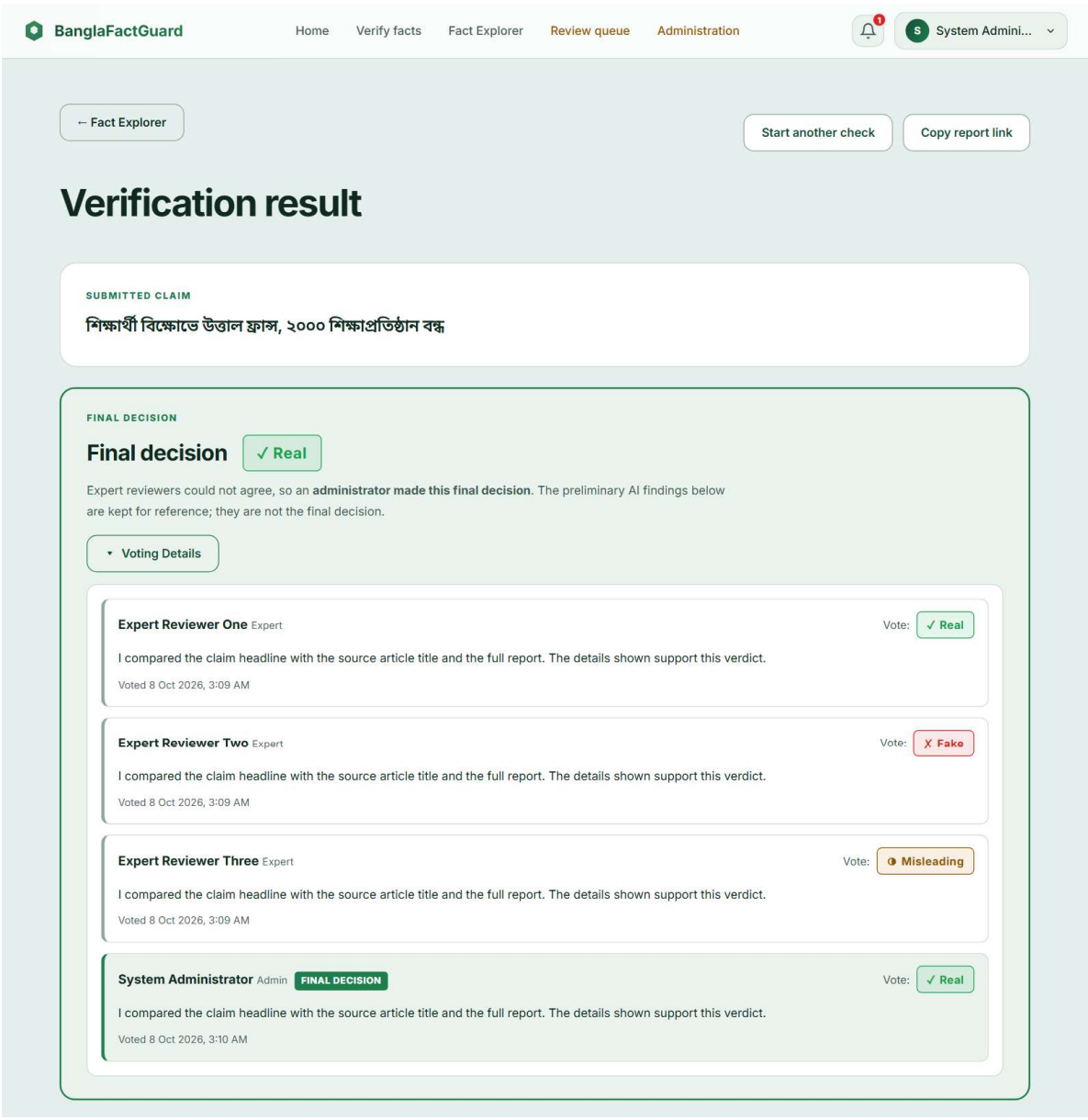


Figure 10.16: Final Decision with Voting Details After an Administrator Decision

10.10 Troubleshooting

Table 10.1: Common Problems and Solutions

Problem	What to do
A publisher is unavailable or unrecognised	Use the registered-source check when available. Ask an administrator to review a publisher that needs registration or activation.
The claim stays “in progress” for a long time	Checks may take a few minutes. The backend must be running with background jobs enabled.
Photo card “temporarily unavailable”	The image-reading service did not return a usable response. Try again later.
The card has no readable headline	Upload a clearer image with the full headline visible. Include the publisher name or logo if available.
Extension cannot capture a page	Click the pinned toolbar icon on that page first; Chrome internal pages cannot be captured, so upload an image instead.
No desktop alert	Check that the extension is ON, notifications are enabled in the Account tab and Chrome may show notifications; Activity still shows the result.

11. Repository Installation and Deployment

11.1 Repository

The project repository is https://github.com/abulkalamazad1425/BanglaFactGuard. Use the following commands to obtain a local copy:

```
git clone https://github.com/abulkalamazad1425/BanglaFactGuard.git
cd BanglaFactGuard
```

11.2 Prerequisites

Table 11.1: Prerequisites

Software	Version / note
Python	3.11 or newer, with PyTorch and model dependencies
Node.js and npm	Node 22 for the website and the extension
PostgreSQL	PostgreSQL database and account; version 17 is listed in the development setup
Docker Compose	For the included Redis and MinIO services
Google Chrome	Desktop Chrome 120 or newer for the extension
Gemini API key	Needed for photo card reading
Multimodal model files	img_backbone.pt, text_backbone.pt, classifier.pt and tokenizer/ in MULTIMODAL_MODEL_DIR

11.3 Backend Setup

Run these PowerShell commands from the repository root:

```
cd backend
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install --upgrade pip
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
.\.venv\Scripts\python.exe -m playwright install chromium
if (!(Test-Path .env)) { Copy-Item .env.example .env }
```

Configure backend/.env for your database, storage and external services (Table 11.2). Keep an existing configuration when updating the project. Use a generated signing secret and keep credentials outside version control.

Table 11.2: Main Configuration Settings (backend/.env)

Setting	Purpose
DB_HOST, DB_PORT, DB_NAME, DB_USER, DB_PASSWORD	PostgreSQL connection
REDIS_HOST, REDIS_PORT	Redis connection
MINIO_*	Image storage endpoint, credentials and bucket
AUTH_SECRET_KEY	Required JWT signing key (at least 32 characters)
ML_*	Language-model configuration for article and headline comparison
GEMINI_API_KEY, GEMINI_*	Photo card reading and request batches
MULTIMODAL_MODEL_DIR, MULTIMODAL_LOAD_ON_STARTUP	Text & image model files and loading
JOBS_ENABLED, JOBS_*	Background worker and recovery
EMAIL_SMTP_*, EMAIL_FROM_ADDRESS, EMAIL_WEBSITE_URL	OTP and final-result emails

11.4 Services, Database and Sources

Create the PostgreSQL database named in DB_NAME. Then, from backend/, start Redis and MinIO, apply the migrations, seed the verified sources and start the API:

```
docker compose up -d
.\.venv\Scripts\python.exe -m alembic upgrade head
.\.venv\Scripts\python.exe scripts/seed_verified_sources.py
.\.venv\Scripts\python.exe run.py
```

The local API is available at http://localhost:8000, with interactive documentation at http://localhost:8000/docs. The migrations include an initial administrator account. Use an appropriately configured administrator account to create expert accounts.

11.5 Website

```
cd frontend
npm ci
npm start
```

Open http://localhost:4200. The development API address is set in frontend/src/environments/environment.ts (default http://localhost:8000/api/v1). A production build is made with npm run build and needs a web server that forwards /api/v1 to FastAPI.

11.6 Browser Extension (Installation)

```
cd extension
npm ci
npm run build
```

1. Open chrome://extensions and turn on **Developer mode**.
2. Click **Load unpacked** and select the extension/dist folder.
3. Pin **BanglaFactGuard — Browser Extension** and click its icon on a news page.

11.7 Running the Tests

```
cd backend;  .\.venv\Scripts\python.exe -m pytest tests/unit -q --no-cov
cd frontend; npx ng test --watch=false --browsers=ChromeHeadless
cd extension; npm test
```

11.8 Deployment Status

- The report documents a local setup with the website, backend, PostgreSQL, Redis, MinIO and an unpacked Chrome extension. The included screenshots illustrate this setup.
- A public production deployment is not documented as completed. Production use would require a configured web server, service addresses, credentials and validation in that environment.

12. Conclusion

BanglaFactGuard was developed to help users check the claims presented in Bangla news photocards. It brings headline extraction, source identification, evidence retrieval and claim comparison into a single workflow. Instead of leaving users to search for related reports manually, the system presents relevant evidence alongside its assessment. This gives users a clearer basis for deciding whether a photocard accurately represents a published news report.

An important distinction in the project is the difference between identifying a publisher and checking a claim. A familiar logo or a matching source name does not, by itself, establish that a headline is accurate. BanglaFactGuard therefore considers the available source, headline and date information separately. When the claimed publisher cannot be identified, the system can search registered active news sources for supporting evidence. Expert review provides a further way to examine cases that need human judgment.

The evaluation also shows why automated results need to be interpreted carefully. The headline comparison model achieved 80% accuracy on the held-out test set. This result supports its use within the verification process, but it does not establish the accuracy of the complete system. A final assessment also depends on whether the image is readable, whether extraction preserves the claim and whether relevant evidence can be retrieved. Missing evidence should therefore be treated as a limitation of the check rather than proof that a claim is false.

The project provides a working foundation for evidence-based verification of Bangla news photocards. Further work should focus on evaluating complete verification outcomes, improving difficult cases and reducing the time required to retrieve and assess evidence. Its practical value lies in helping users examine a claim with supporting information while making the limits of the automated assessment clear.

13. References

[1] F. T. J. Faria, M. B. Moin, Z. Hasan, Md. A. A. Khandaker, N. Islam, K. M. Hasib and M. F. Mridha, "MultiBanFakeDetect: Integrating advanced fusion techniques for multimodal detection of Bangla fake news in under-resourced contexts," *International Journal of Information Management Data Insights*, vol. 5, Art. no. 100347, 2025.

[2] A. Bhattacharjee, T. Hasan, W. U. Ahmad, K. S. Mubasshir, M. S. Islam, A. Iqbal, M. S. Rahman and R. Shahriyar, "BanglaBERT: Language Model Pretraining and Benchmarks for Low-Resource Language Understanding Evaluation in Bangla," in *Findings of the Association for Computational Linguistics: NAACL 2022*, 2022, pp. 1318–1327.

[3] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in *Proc. 36th International Conference on Machine Learning (ICML)*, 2019, pp. 6105–6114.

[4] F. Feng, Y. Yang, D. Cer, N. Arivazhagan and W. Wang, "Language-agnostic BERT Sentence Embedding," in *Proc. 60th Annual Meeting of the Association for Computational Linguistics*, 2022, pp. 878–891.

[5] L. Bonifacio, V. Jeronymo, H. Q. Abonizio, I. Campiotti, M. Fadaee, R. Lotufo and R. Nogueira, "mMARCO: A Multilingual Version of the MS MARCO Passage Ranking Dataset," arXiv:2108.13897, 2021.

[6] P. He, J. Gao and W. Chen, "DeBERTaV3: Improving DeBERTa using ELECTRA-Style Pre-Training with Gradient-Disentangled Embedding Sharing," arXiv:2111.09543, 2021.

[7] R. Panchendrarajan and A. Zubiaga, "Claim detection for automated fact-checking: A survey on monolingual, multilingual and cross-lingual research," *Natural Language Processing Journal*, vol. 7, Art. no. 100066, 2024.

[8] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed. New York: McGraw-Hill, 2020.